SIC - MMAB: Synchronisation Involves Communication in Multiplayer Multi-Armed Bandits

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Abstract

Motivated by cognitive radio networks, we consider the stochastic multiplayer multi-armed bandit problem, where several players pull arms simultaneously and collisions occur if one of them is pulled by several players at the same stage. We construct a decentralized algorithm that achieves the same performance as a centralized one as long as players are synchronized and observe their collisions. We actually achieve this result by constructing a communication protocol between players that deliberately enforces collisions, allowing them to share their information at a negligible cost.

Under a weaker assumption on the feedback, i.e., collisions are not observed, we still maintain some communication between players but at the cost of some extra multiplicative term in the regret.

Finally, we prove that the logarithmic growth of the regret is still achievable in the dynamic case where players are not synchronized with each other, thus preventing communication.

1 Introduction

In the stochastic Multi Armed Bandit problem (MAB), a single player sequentially takes a decision (or "pulls an arm") amongst a finite set of possibilities $[K] := \{1, \dots, K\}$. After pulling arm $k \in [K]$ at stage $t \in \mathbb{N}$, the player receives a random reward $X_k(t) \in [0, 1]$, drawn *i.i.d.* according to some unknown distribution ν_k of expectation $\mu_k := \mathbb{E}[X_k(t)]$. Her objective is to maximize her cumulative reward up to stage $T \in \mathbb{N}$. This sequential decision problem, first introduced for clinical trials [Thompson, 1933, Robbins, 1952], involves an "*exploration/exploitation* dilemma" where the player must balance acquiring *vs.* using information. The performance of an algorithm is controlled in term of **regret**, the difference of the cumulated reward of an optimal algorithm knowing the distributions $(\nu_k)_{k\in[K]}$ beforehand and the cumulated reward of the player. It is known that any "reasonable" algorithm must incur at least a logarithmic regret [Lai and Robbins, 1985], which is attained by some existing algorithms such as UCB [Agrawal, 1995, Auer et al., 2002].

MAB has been recently popularized thanks to its applications to online recommendation systems. Many different variants of MAB and classes of algorithms have thus emerged in the recent

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years [see Bubeck and Cesa-Bianchi, 2012]. In particular, they have been considered for cognitive radios [Jouini et al., 2009], where the problem gets more intricate as multiple users are involved and they collide if they pull the same arm k at the same time t, i.e., they transmit on the same channel. If this happens, they all receive 0 instead of $X_k(t)$, meaning that no message is transmitted.

If a central agent were to control simultaneously all players' behavior then a tight lower bound is known [Anantharam et al., 1987, Komiyama et al., 2015]. Yet this centralized problem is not adapted to cognitive radios, as it allows communication between players at each time step, which in practice represents significant costs in both energy and time. As a consequence, most of the current interest lies in the decentralized case [Liu and Zhao, 2010, Anandkumar et al., 2011, Avner and Mannor, 2014], which presents another complication due to the feedback. Besides the reward of the arm, an additional piece of information may be observed.

When this extra observation is the collision indicator, Rosenski et al. [2016] provided two algorithms for both a fixed and a varying number of players. They are based on a *Musical Chairs* procedure that quickly assigns players to different arms. Besson and Kaufmann [2018] provided an efficient UCB-based algorithm if $X_k(t)$ is observed instead. Lugosi and Mehrabian [2018] recently proposed an algorithm using no additional information. The performances of these algorithms and the underlying model differences are summarized in Table 1, Section 1.1.

The first non trivial lower bound for this problem has been recently improved [Liu and Zhao, 2010, Besson and Kaufmann, 2018]. These lower bounds suggest that decentralization adds to the regret a multiplicative factor M, the number of players, compared to the centralized case [Anantharam et al., 1987].

Interestingly, these lower bounds scale linearly with the inverse of the gaps between the μ_k whereas this scaling is quadratic for most of the existing algorithms. This is due to the fact that although collisions account for most of the regret, lower bounds are proved without considering them.

Although it is out of our scope, a variant of the multiplayer bandits introduced by Kalathil et al. [2014] is worth mentioning. Indeed, under the hypothesis that the reward distribution depends on each user, an objective is to find a Pareto optimal allocation of the arms. Such a configuration was reached at the cost of restricted communication between the players [Avner and Mannor, 2015, 2018]. Bistritz and Leshem [2018] recently proposed an algorithm without explicit communication, but relying on synchronization between the players.

Our main contributions are the following:

Section 2: When collisions are observed, we propose a decentralized algorithm allowing players to communicate statistics and additional information through deliberate collisions. The regret of this algorithm reaches (up to a multiplicative universal constant) the lower bound of the centralized problem, meaning that the aforementioned lower bounds are unfortunately incorrect.

Section 3: Without observing collisions, we still recover a logarithmic growth of regret under an additional mild assumption. The cost is basically a multiplicative factor M.

These first two algorithms both rely on the unrealistic assumption that all users start transmitting at the very same time. It also explains why the current literature fails to provide near optimal results for the multiplayer bandits. It therefore appears that the assumption of synchronization has to be removed for practical application of the multiplayer bandits problem.

Section 4: Without synchronization, we construct another algorithm with logarithmic regret, yet the dependencies in the gaps between rewards become quadratic.

1.1 Models

In this section, we introduce the different models of multiplayer MAB with a known number of arms K but an unknown number of players M. At each time step $t \in [T]$, given their (private) information, all players $j \in [M]$ simultaneously pull the arms $\pi^j(t)$ and receive the binary reward

$$r^j(t) := X_{\pi^j(t)}(t)(1 - \eta_{\pi^j(t)}(t)) \text{ where } \eta_{\pi^j(t)}(t) \text{ is the collision indicator defined by:}$$
$$\eta_k(t) := \mathbb{1}_{\#C_k(t)>1} \qquad \text{with} \qquad C_k(t) := \{j \in [M] \mid \pi^j(t) = k\}.$$

The problem is **centralized** if players can communicate any information to each other. In that case, they can easily avoid collisions and share their statistics. In opposition, the problem is **decentralized** when players have only access to their own rewards and actions. The crucial concept we introduce is (a)synchronization between players.

Assumption 1 (Synchronization)**.** *Player i enters the bandit game at the time $\tau_i = 0$ and stays until the final horizon T. This is common knowledge to all players.*

With this assumption, the model is called **static**.

Assumption 2 (Asynchronization)**.** *Players enter at different times $\tau_i \in [T]$ and stay until the final horizon T. The τ_i are unknown to all players (including i).*

With asynchronicity, the model is **dynamic** and several variants already exist [Rosenski et al., 2016]. Denote by $\mathbf{M}(t)$ the set of players in the game at time t (unknown but not random) and by $\mu_{(n)}$ the n-th order statistics of μ, i.e., $\mu_{(1)} \geq \mu_{(2)} \geq \ldots \geq \mu_{(K)}$. The total regret is then defined for both models by:

$$R_T := \sum_{t=1}^{T} \sum_{k=1}^{\#\mathbf{M}(t)} \mu_{(k)} - \mathbb{E}_\mu\Big[\sum_{t=1}^{T} \sum_{j \in \mathbf{M}(t)} r^j(t)\Big]$$

As mentioned in the introduction, different observation settings will be considered.

Collision Sensing: Player j observes $\eta_{\pi^j(t)}(t)$ and $r^j(t)$ at each time step.

No sensing: Player j only observes $r^j(t)$, i.e., a reward of 0 can indistinguishably come from a collision with another player or a null statistic $X_{\pi^j(t)}(t)$.

As a remark, the setting, where both $X_{\pi^j}(t)$ and $r^j(t)$ are observed, is also often considered in the literature and is called **Statistic Sensing** [Besson and Kaufmann, 2018]. The No Sensing setting is the most difficult as there is no extra observation.

Table 1 below compares the performances of the major algorithms, specifying the precise setting considered for each of them. Our algorithms also have problem independent bounds that are not mentioned in Table 1 for the sake of clarity. In order to have precise bounds, we first introduce $\bar{\Delta}_{(M)} := \min_{i=1,\ldots,M}(\mu_{(i)} - \mu_{(i+1)})$ the smallest gap amongst the top-$M+1$ arms, and $\Delta' := \min\{\mu_{(M)} - \mu_i \mid \mu_{(M)} - \mu_i > 0\}$. Note that the two dynamic algorithms in Table 1 rely on different specific assumptions.

2 Collision Sensing: Achieving Centralized Performances by Communicating through Collisions

In this section, we consider the Collision Sensing static model and prove that the decentralized problem is almost as complex, in terms of regret growth, as the centralized one. When players are

Model	Algorithm	Known values	Upper bound (up to universal constant factor)
Centralized Multiplayer	MP-TS	M	$\sum_{k>M} \frac{\log(T)}{\mu_{(M)}-\mu_{(k)}}$
Decentralized, Statistic Sensing	Besson and Kaufmann [2018]	M	$M^3 \sum_{1\leq i<k\leq K} \frac{\log(T)}{(\mu_{(i)}-\mu_{(k)})^2}$
Decentralized, Collision Sensing	Rosenski et al. [2016]	$\mu_{(M)} - \mu_{(M+1)}$	$\frac{MK\log(T)}{(\mu_{(M)}-\mu_{(M+1)})^2}$
Decentralized, Collision Sensing	SIC-MMAB	-	$\sum_{k>M} \frac{\log(T)}{\mu_{(M)}-\mu_{(k)}} + KM\log(T)$
Decentralized, No Sensing	Lugosi and Mehrabian [2018]	M	$\frac{KM}{(\mu_{(M)}-\mu_{(M+1)})^2}\log(T)$
Decentralized, No Sensing	Lugosi and Mehrabian [2018]	$M, \mu_{(M)}$	$\frac{MK^2}{\mu_{(M)}}\log^2(T)$ $+KM\min(\sqrt{T\log(T)}, \frac{\log(T)}{\Delta'})$
Decentralized, No Sensing	SIC-MMAB'	$\mu_{(K)}$	$\sum_{k>M} \frac{\log(T)}{\mu_{(M)}-\mu_{(k)}}$ $+\frac{KM^3\log(T)}{\mu_{(K)}}\log^2\left(\frac{\log(T)}{(\mu_{(M)}-\mu_{(M+1)})^2}\right)$
Decentralized, No Sensing	SIC-MMAB2	$\mu_{(K)}$	$M\sum_{k>M} \frac{\log(T)}{\mu_{(M)}-\mu_{(k)}} + \frac{MK^2}{\mu_{(K)}}\log(T)$
Decentralized, Collision Sensing , Dynamic	Rosenski et al. [2016]	$\bar{\Delta}_{(M)}$	$\frac{M\sqrt{K\log(T)T}}{\bar{\Delta}_{(M)}}$
Decentralized, No Sensing, Dynamic	DYN-MMAB	-	$\frac{MK\log(T)}{\bar{\Delta}^2_{(M)}} + \frac{M^2K\log(T)}{\mu_{(M)}}$

Table 1: Performances of different algorithms. Our algorithms and results are highlighted in red.

synchronized, we provide an algorithm reaching (up to a multiplicative universal constant) the lower bound of the centralized multiplayer MAB [Anantharam et al., 1987]. This algorithm strongly relies on the synchronization assumption, which we leverage to allow communication between players through observed collisions. The communication protocol is detailed and explained in Section 2.2.3.

This result also implies that the two lower bounds provided in the literature [Besson and Kaufmann, 2018, Liu and Zhao, 2010] are unfortunately not correct. Indeed, the factor M that was supposed to be the cost of the decentralization in the regret should not appear.

Let us now describe our algorithm SIC-MMAB. It consists in several phases:

1. First, an initialization phase estimates the number of players and assigns a rank to each player.
2. Players then alternate between exploration phases and communication phases
   (a) During the exploration phases, the performance of each arm is estimated in a Successive Accepts and Rejects fashion [Perchet and Rigollet, 2013, Bubeck et al., 2013].
   (b) During the communication phases, players share their statistics using collisions. At the end of such a phase, all statistics are updated and shared amongst players.
3. The last phase, the exploitation one, is triggered as soon as an arm is detected as optimal and consists in pulling this arm until the final horizon T.

2.1 Some preliminary notations

For the sake of clarity, let us first introduce some notations. Players who are not in the exploitation phase are called **active**. We denote, with a slight abuse of notation, by $[M_p]$ the set of active players during the p-th phase of exploration-communication and by $M_p \leq M$ its cardinality. Notice that M_p is non increasing because players never leave the exploitation phase.

Any arm among the top-M ones is called **optimal** and any other arm is **sub-optimal**. Arms that still need to be explored (players cannot determine whether they are optimal or not yet) are **active**. We denote, with the same abuse of notation, the set of active arms by $[K_p]$ of cardinality $K_p \leq K$. By construction of our algorithm, this set is common to all players at each stage.

Our algorithm is based on a protocol called *sequential hopping* [Joshi et al., 2018]. It consists in incrementing the index of the arm pulled by a specific player: if she plays arm π_t^k at time t, she will play $\pi_{t+1}^k = \pi_t^k + 1 \pmod{[K_p]}$ at time $t+1$ during phase p.

In what follows, the statement *"with proba $1 - \mathcal{O}(\delta(T))$, it holds that $f(T) = \mathcal{O}(g(T))$"* means that there is a universal constant $c \in \mathbb{R}_+$ such that $f(T) \leq cg(T)$ with probability at least $1 - c\delta(T)$.

2.2 Description of our protocol

As mentioned before, the SIC-MMAB algorithm consists in several phases. At the end of a communication phase, players update and share the same statistics, so that the decentralized case is actually similar to the centralized one. After alternating enough times between exploration and communication phases, sub-optimal arms are eliminated and players are fixed to different optimal arms and will exploit them until stage T. The pseudocode of SIC-MMAB is given in Algorithm 1, Appendix A.1.

Notice that SIC-MMAB can be easily adapted to be "fair" among players (they asymptotically have the same average reward): during the exploitation phase, players just have to sequentially hop among the optimal arms. Similarly, exploration phases can be made by mini-batches to reduce the number of switches [Perchet et al., 2015]. Each phase is described more thoroughly below.

2.2.1 Initialization phase

The objective of the first phase is to estimate the number of players M and to assign **internal ranks** amongst players. First, players follow the Musical Chairs algorithm, described in Pseudocode 3 (see Appendix A.1), during T_0 steps in order to reach an **orthogonal setting**, i.e., a position where they are all pulling different arms. The index of the arm pulled by a player at stage T_0 then corresponds to her **external rank**.

The second procedure, given by Pseudocode 4 and illustrated in Figure 3 in Appendix A.1, determines M and assigns a unique internal rank in $[M]$ to each player. For example, if there are three players on arms 5, 7 and 2 at $t = T_0$, their external ranks respectively correspond to 5, 7 and 2 while their internal ranks are 2, 3 and 1. Roughly speaking, the players follow each other sequentially hopping through all the arms. Each player then deduces M and her internal rank from observed collisions during this procedure that lasts $2K$ steps.

During the other phases, active players will always know the set of active players $[M_p]$ and each of them has a unique internal rank in $[M_p]$. This is how the initial symmetry among players is broken and it allows the decentralized algorithm to establish communication protocols.

2.2.2 Exploration phase

The p-th exploration phase consists in sequential hopping among the active arms for $K_p 2^p$ steps. Any active arm is thus pulled 2^p times by each active player. Using their internal rank, players start and remain in an orthogonal setting during the exploration phase, which is collision-less.

We denote by $B_s = \sqrt{\frac{2\log(T)}{s}}$ the standard error bound after s pulls and by $T_k(p)$ (resp. $S_k(p)$) the centralized total number of pulls (resp. sum of rewards) for the arm k during the p first explo-

ration phases, i.e., $T_k(p) = \sum_{j=1}^{M} T_k^j(p)$ where $T_k^j(p)$ is the number of pulls for the arm k by player j during the p first exploration phases.

After a succession of two phases (exploration and communication), an arm k is **accepted** if

$$\#\Big\{ i \in [K_p] \,\Big|\, \hat{\mu}_k(p) - B_{T_k(p)} \geq \hat{\mu}_i(p) + B_{T_i(p)} \Big\} \geq K_p - M_p,$$

where $\hat{\mu}_k(p) = \frac{S_k(p)}{T_k(p)}$ is the centralized empirical mean of the arm k. This inequality implies that at least $K_p - M_p$ arms among the active arms are worse than k with high probability.

In the same way, k is **rejected** if

$$\#\Big\{ i \in [K_p] \,\Big|\, \hat{\mu}_i(p) - B_{T_i(p)} \geq \hat{\mu}_k(p) + B_{T_k(p)} \Big\} \geq M_p,$$

meaning that there are at least M_p active arms better than k with high probability.

2.2.3 Communication phase

In this phase, each active player communicates, one at a time, her statistics on the active arms to all other active players. Recall that since an arm is pulled 2^p times by a single player during the p-th exploration phase, the statistics of an arm can be sent by a player in $p+1$ bits, i.e., at most $p+1$ collisions. An active player can have three possible statuses during this phase:

1. either she is receiving some other players' information about the arm k. In that case, she proceeds to **Receive Protocol** (see Pseudocode 1) to learn a total reward observed by another player on k during the previous exploration phase.

2. Or she is sending her statistics about arm k to player l (who is then receiving statistics). In that case, she proceeds to **Send Protocol** (see Pseudocode 2) to send the total observed reward on k during the previous exploration phase in a time $p+1$.

3. Or she is pulling the arm corresponding to her rank, while waiting for other players to finish communicating statistics among them.

Communicated statistics are all of length $p+1$, even if they could be sent with shorter messages, in order to keep synchronization among players. Since all active players are synchronized and know $[M_p]$, $[K_p]$ and their internal rank, the coordination of all these statuses among them is possible. At the end of the communication phase, all active players share the same information. They know which arms must be accepted or rejected. Rejected arms are removed right away from the set of active arms. Thanks to the assigned ranks, accepted arms are assigned to one player each. The remaining active players then update both sets of active players and arms as well as their rank as described in Algorithm 1, line 26.

2.3 Regret analysis

This section is devoted to the analysis of our algorithm. We decompose the regret as follows:

$$R_T = R^{\text{init}} + R^{\text{comm}} + R^{\text{explo}} \tag{1}$$

Receive Protocol

Input: p (phase number), j (own internal rank), $[K_p]$ (set of active arms)

$s \leftarrow 0$
$\pi \leftarrow$ index of the j-th active arm
for $n = 0, \dots, p$ **do**
  Pull π
  if $\eta_\pi(t) = 1$ **then** *# other player sends* 1
  $s \leftarrow s + 2^n$ **end if**
end for
return s *# statistics for arm* k

Pseudocode 1: receive statistics of length $p+1$

Send Protocol

Input: l (player receiving), s (statistics to send), p (phase number), j (own internal rank), $[K_p]$ (set of active arms)

$m \leftarrow$ binary writing of s of length $p+1$, i.e., $s = \sum_{n=0}^{p} m[n]2^n$
for $n = 0, \dots, p$ **do**
  if $m[n] = 1$ **then**
    Pull the l-th active arm *# send* 1
  else Pull the j-th active arm *# send* 0
  end if
end for

Pseudocode 2: send statistics s of length $p+1$ to player l

$$\text{where} \begin{cases} R^{\text{init}} = T_{\text{init}} \sum_{k=1}^{M} \mu_{(k)} - \mathbb{E}_\mu\big[\sum_{t=1}^{T_{\text{init}}} \sum_{j=1}^{M} r^j(t)\big] \text{ with } T_{\text{init}} = T_0 + 2K \\ R^{\text{comm}} = \mathbb{E}_\mu\big[\sum_{t \in \text{Comm}} \sum_{j=1}^{M} (\mu_{(j)} - r^j(t))\big] \text{ with Comm the set of communication steps} \\ R^{\text{explo}} = \mathbb{E}_\mu\big[\sum_{t \in \text{Explo}} \sum_{j=1}^{M} (\mu_{(j)} - r^j(t))\big] \text{ with Explo} = \{T_{\text{init}} + 1, \dots, T\} \setminus \text{Comm} \end{cases}$$

A **communication step** is defined as a time step where at least one player is communicating statistics, i.e., using **Send Protocol**. These terms respectively correspond to the regret due to the initialization phase, the communication and the regret of both exploration and exploitation phases.

2.3.1 Initialization analysis

The initialization regret is obviously bounded by $M(T_0 + 2K)$. Lemma 1 provides the probability to reach an orthogonal setting at time T_0 and its proof is delayed to Appendix A.2.1. If this orthogonal setting is reached, the initialization phase is **successful**.

Lemma 1. *After time T_0, all players pull different arms with probability at least $1 - M \exp\left(-\frac{T_0}{K}\right)$.*

2.3.2 Exploration regret

For both exploration and exploitation phases, we control the number of times an arm is pulled before being accepted or rejected. The proof is delayed to Appendix A.2.2.

Proposition 1. *With probability $1 - \mathcal{O}\left(\frac{K \log(T)}{T} + M \exp\left(-\frac{T_0}{K}\right)\right)$, every optimal arm k is accepted after at most $\mathcal{O}\left(\frac{\log(T)}{(\mu_k - \mu_{(M+1)})^2}\right)$ pulls during exploration phases, and every sub-optimal arm k is rejected after at most $\mathcal{O}\left(\frac{\log(T)}{(\mu_{(M)} - \mu_k)^2}\right)$ pulls during exploration phases.*

For both exploration and exploitation phases, the decomposition used in the centralized case [Anantharam et al., 1987] holds because there are absolutely no collisions during these two types of phases (at least conditionally to the success of the initialization phase):

$$R^{\text{explo}} = \sum_{k>M} (\mu_{(M)} - \mu_{(k)}) T_{(k)}^{\text{explo}}(T) + \sum_{k\leq M} (\mu_{(k)} - \mu_{(M)})(T^{\text{explo}} - T_{(k)}^{\text{explo}}(T))$$

Where $T^{\text{explo}} = \#\text{Explo}$ and $T_{(k)}^{\text{explo}}(T)$ is the (centralized) number of time steps where the k-th best arm is pulled in these two types of phases.

Lemma 2. *With probability* $1 - \mathcal{O}\left(\frac{K\log(T)}{T} + M\exp\left(-\frac{T_0}{K}\right)\right)$*, all the following holds simultaneously:*

i) for a suboptimal arm k, $\quad T_k^{\text{explo}}(T) = \mathcal{O}\left(\min\left\{\frac{\log(T)}{(\mu_{(M)}-\mu_k)^2},\ T\right\}\right)$

ii) $\sum_{k\leq M} (\mu_{(k)} - \mu_{(M)})(T^{\text{explo}} - T_{(k)}^{\text{explo}}) = \mathcal{O}\left(\sum_{k>M} \min\left\{\frac{\log(T)}{\mu_{(M)}-\mu_{(k)}}, \sqrt{T\log(T)}\right\}\right)$

iii) $\quad R^{\text{explo}} = \mathcal{O}\left(\sum_{k>M} \min\left\{\frac{\log(T)}{\mu_{(M)}-\mu_{(k)}}, \sqrt{T\log(T)}\right\}\right)$

The first part is a direct consequence of Proposition 1. The proof of the second part is also delayed to Appendix A.2.3 and the last part immediately follows.

2.3.3 Communication cost

We now focus on the R^{comm} term in Equation (1). Lemma 3 states it is negligible compared to $\log(T)$ and has a significant impact on the regret only for small values of T.

Lemma 3. *With probability* $1 - \mathcal{O}\left(\frac{K\log(T)}{T} + M\exp\left(-\frac{T_0}{K}\right)\right)$*, the following holds:*

$$R^{\text{comm}} = \mathcal{O}\left(KM^3\log^2\left(\min\left\{\frac{\log(T)}{(\mu_{(M)} - \mu_{(M+1)})^2},\ T\right\}\right)\right).$$

Proof. As explained in Section 2.2.3, the length of the communication phase $p \in [N]$ is at most $KM^2(p+1)$. Conditionally to the successes of the initialization and the estimations of the arms as in Proposition 1, the number of exploration phases N is at most $\mathcal{O}\Big(\log\left(\min\left\{\frac{\log(T)}{(\mu_{(M)}-\mu_{(M+1)})^2},\ T\right\}\right)\Big)$, so the cost of communication is smaller than $KM^3\sum_{p=1}^{N}(p+1) \leq \mathcal{O}\left(KM^3N^2\right)$. □

2.3.4 Total regret

Theorem 1, deduced from Lemmas 1, 2, 3 and Equation (1), finally bounds the regret of SIC-MMAB.

Theorem 1. *With the choice* $T_0 = \lceil K\log(T)\rceil$*, for any given set of parameters* K*,* M *and* $\boldsymbol{\mu}$*:*

$$\begin{aligned}\mathbb{E}\big[R_T\big] \leq & c_1 \sum_{k>M} \min\left\{\frac{\log(T)}{\mu_{(M)} - \mu_{(k)}}, \sqrt{T\log(T)}\right\} + c_2 KM\log(T) \\ & + c_3 KM^3\log^2\left(\min\left\{\frac{\log(T)}{(\mu_{(M)} - \mu_{(M+1)})^2}, T\right\}\right)\end{aligned}$$

where c_1*,* c_2 *and* c_3 *are universal constants.*

2.4 In contradiction with existing lower bounds ?

Theorem 1 is in contradiction with the two existing lower bounds [Besson and Kaufmann, 2018, Liu and Zhao, 2010], however SIC-MMAB respects the conditions required for both. It was thought that the cost of the decentralization was a multiplicative factor M compared to the centralized lower bound [Anantharam et al., 1987]. However, it appears that the regret of the decentralized case is not that much different from the one in the centralized case, at least if players are synchronized. Indeed, SIC-MMAB actually takes advantage of this synchronization to establish communication protocols as players are able to communicate through collisions.

The incorrect statement of Besson and Kaufmann [2018] seems to be found in Lemma 12. It claims that the probability distribution of collisions does not depend on the distributions of the arms when it does, especially for SIC-MMAB. The mistake of Liu and Zhao [2010] would be in Lemma 3, since the proof omits the fact that the actions of a player also depend on the previous collisions.

Our algorithm reaches, up to a constant factor, the lower bound of the centralized problem, up to an additional term $\mathcal{O}(KM\log(T))$ due to the initialization. If a pre-agreement is allowed among players, this phase would not be needed and this additional term would also not appear. As a consequence, SIC-MMAB shows that the decentralized multiplayer bandits with synchronization is almost equivalent to the centralized multiplayer problem which seems *a priori* much simpler.

As synchronization is not a reasonable assumption for practical purposes, and that it leads to undesirable algorithms, this assumption should be removed in the multiplayer MAB. This is what we called the *dynamic model*. However, this problem seems complex to model formally, as there must be restrictions on how players can enter or leave the game, or find an appropriate alternative formulation of the regret. Indeed, if players can enter and leave too frequently, then no learning can occur and the regret would be linear. The difficulty to have an interesting, nontrivial and formal model under this setting explains why most of the literature focused on the synchronized model.

2.5 Experiments

We now compare the empirical performances of SIC-MMAB with the MCTopM algorithm [Besson and Kaufmann, 2018], on generated data. We also compared with the MusicalChairs algorithm [Rosenski et al., 2016], but its performance was irrelevant and out of scale. All the considered regret values are averaged over 200 runs.

Figure 1 represents the evolution of the regret for both algorithms with the following problem parameters: $K = 9$, $M = 6$, $T = 5 \times 10^5$. The means of the arms are linearly distributed between 0.9 and 0.89, so the gap between two consecutive arms is 1.25×10^{-3}. The switches between exploration phases and communication phases for SIC-MMAB are easily observable.

Figure 2 represents the evolution of the final regret as a function of the gap Δ between two consecutive arms in a logarithmic scale. The problem parameters K, M and T are the same. Although MCTopM seems to provide better results with larger values of Δ, SIC-MMAB seems to have a smaller dependency in $1/\Delta$. This confirms the theoretical results claiming that MCTopM scales with Δ^{-2} while SIC-MMAB scales with Δ^{-1}. This can be observed on the left part of Figure 2 where the slope for MCTopM is approximately twice as big as for SIC-MMAB. Also, a different behavior of the regret appears for very low values of Δ which is certainly due to the fact that the regret is constant for extremely small values of Δ.

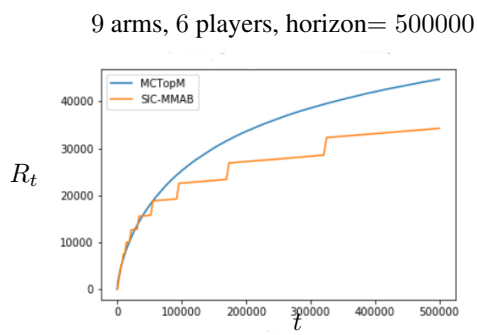

Figure 1: Evolution of regret over time

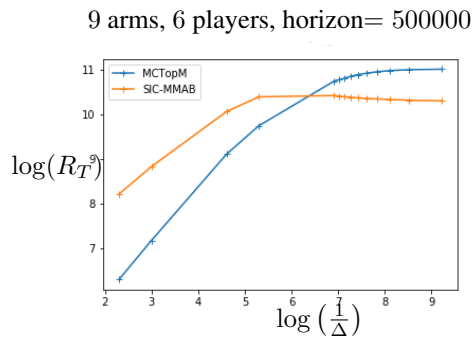

Figure 2: Final regret as a function of $\frac{1}{\Delta}$ (logarithmic scale)

3 No Sensing: Communication through Synchronization

This section focuses on the No Sensing setting but still with synchronization. First of all, we claim that a communication protocol similar to the one of SIC-MMAB can be devised here, under a mild extra assumption: a lower bound $\mu_{\min}$ of the average rewards μ_k is known. Indeed, in the Collision Sensing model, a bit is sent through a single collision. Without sensing, it can be done with arbitrary high probability using $\frac{\log(T)}{\mu_{\min}}$ time steps. This only adds a multiplicative factor of $\frac{\log(T)}{\mu_{\min}}$ to the communication regret. So, SIC-MMAB can be easily adapted for the No Sensing model into the SIC-MMAB' algorithm with a regret scaling as:

$$\mathcal{O}\left(\sum_{k>M} \frac{\log(T)}{\mu_{(M)} - \mu_{(k)}} + \frac{KM^3 \log(T)}{\mu_{\min}} \log^2\left(\frac{\log(T)}{(\mu_{(M)} - \mu_{(M+1)})^2}\right)\right)$$

Although this regret is not asymptotically logarithmic (because of the $\log(T)\log^2(\log(T))$ term), for specific problem parameters and for a fixed horizon T, this upper bound would be lower than the natural regret bound without communication between players: $\sum_{k>M} \frac{M\log(T)}{\mu_{(M)}-\mu_{(k)}}$.

This remark illustrates how complex an anytime lower bound for the No Sensing setting can be. In this section, we introduce an algorithm for the synchronized case. It also relies on a communication protocol, but with more limited information, which thus incurs a much better dependency in M as well as a logarithmic regret.

Simple algorithms such as UCB seem to perform well (on generated data) but appear to incur a linear regret with some constant probability in the No Sensing setting [Besson and Kaufmann, 2018]. In Appendix D, the discussion about UCB is extended and some reasons of its failure are explained for the No Sensing model, using algebraic arguments (Lindemann-Weierstrass Theorem).

3.1 Adapted Communication Protocol

The algorithm SIC-MMAB2 is formally described in Appendix B.1. It relies on several subroutines that are detailed in the appendix. Similarly to SIC-MMAB, it starts with an initialization phase to estimate M. It then alternates between exploration and communication phases, but the goal of the communication phases is to communicate to other players that an arm is detected as optimal or sub-optimal (instead of transmitting statistics). This allows to share common sets of active arms and

players. Protocols to declare such arms and to detect declarations from other players are detailed in Appendix B.1.3. The algorithm then ends with an exploitation phase as for SIC-MMAB.

An additional assumption is required for SIC-MMAB2. This assumption is quite similar to an assumption made by Lugosi and Mehrabian [2018] for the No Sensing model.

Assumption 3. *A lower bound of $\mu_{(K)}$ is known to all players:* $0 < \mu_{\min} \leq \min_i \mu_i$

The regret incurred by SIC-MMAB2 is given by Theorem 2. Its proof is delayed to Appendix B.2.

Theorem 2. *With the choice $T_c = \lceil \frac{\log(T)}{\mu_{\min}} \rceil$ for the initialization,* SIC-MMAB2 *has a regret scaling as*

$$\mathbb{E}[R_T] = \mathcal{O}\left(\sum_{k>M} \min\left\{ \frac{M\log(T)}{\mu_{(M)} - \mu_{(k)}}, \sqrt{T\log(T)} \right\} + \frac{MK^2}{\mu_{\min}} \log(T) \right).$$

4 Without Synchronization, the Dynamic Setting

From now on, we no longer assume that players can communicate using synchronization. In the previous sections, it was crucial that all exploration/communication phases start and end at the same time. This assumption is clearly non-realistic and should be alleviated, as radios do not start and end transmitting simultaneously. Additional assumptions on how players can enter and leave the game are required to avoid a trivial linear regret bound.

We assume for simplicity that players do not leave the game once they have started. Yet, we mention that our results can also be adapted to the cases when players can leave the game during specific intervals or share an internal synchronized clock [Rosenski et al., 2016].

Assumption 2 holds in the whole section: at each stage $t = t_j + \tau_j$, player j does not know t but only t_j (duration since joining). We denote $T^j = T - \tau_j$ the (known) time horizon of player j.

4.1 A Communication-less Protocol

As synchronization is no longer possible, DYN-MMAB algorithm uses different procedures than the previous algorithms. DYN-MMAB is described in Appendix C.1.

The protocol adapted to the dynamic setting is rather intuitive as a player will only follow two different sampling strategies: either she samples uniformly at random in $[K]$; or she exploits an arm and pulls it until the final horizon. In the first case, the exploration of the other players is not too disturbed by collisions as they only change the mean reward of all arms by a roughly constant factor. In the second case, the exploited arm will appear as sub-optimal to the other players, which is actually convenient as this arm is now occupied.

In the first phase of the algorithm, players uniformly explore all the arms, without knowing the number of players in the game (it may evolve dynamically). As soon as a player distinguishes an arm as optimal (i.e., the best available arm), she tries to occupy it as follows: she continues to sample uniformly and occupies this arm as soon as she encounters a positive reward[1]. The player either succeeds to occupy an arm within a given number of stages, and then exploits it until the final horizon, or she fails, then marks this arm as occupied and looks for the next available optimal arm.

The regret incurred by DYN-MMAB in the dynamic No Sensing model is given by Theorem 3. Its proof is delayed to Appendix C.2. Recall that we assume $\mathbf{M}(t)$ non decreasing in term of inclusion

[1]The naive choice to sample only this arm until observing a positive reward and estimating it already occupied if too many 0 are observed does not deal with cases where several players try to occupy the same arm at the same time.

and the cumulative regret is defined by:

$$R_T = \sum_{t=1}^{T} \sum_{k=1}^{\#\mathbf{M}(t)} \mu_{(k)} - \sum_{t=1}^{T} \sum_{j \in \mathbf{M}(t)} \mu_{\pi^j(t)}(t)(1 - \eta_{\pi^j(t)}(t)).$$

Theorem 3. *In the dynamic setting, the regret incurred by* DYN-MMAB *is upper bounded as follows:*

$$\mathbb{E}[R_T] \leq \mathcal{O}\left(\frac{M^2 K \log(T)}{\mu_{(M)}} + \frac{MK\log(T)}{\bar{\Delta}^2_{(M)}}\right)$$

where $M = \#\mathbf{M}(T)$ is the total number of players in the game and $\bar{\Delta}_{(M)} = \min_{i=1,\dots,M}(\mu_{(i)} - \mu_{(i+1)})$.

In case of equal means between the arms, i.e., $\bar{\Delta}_{(M)} = 0$, an early stopping in the exploration can be set at a time $\tilde{\mathcal{O}}((T-\tau_j)^{\frac{2}{3}})$, to guarantee a regret in $\tilde{\mathcal{O}}(T^{\frac{2}{3}})$, where $\tilde{\mathcal{O}}$ hides polynomial factors in M, K and $\log(T)$.

We mention that Algorithm 3 also leads to a Pareto optimal configuration in the more general problem where users' reward distributions differ [Kalathil et al., 2014, Avner and Mannor, 2015, 2018, Bistritz and Leshem, 2018].

5 Conclusion

We have presented three algorithms for different multiplayer bandits models. The first one illustrates why the assumption of synchronization between the players is basically equivalent to allowing communication. Since it is possible to communicate through collisions with other players, the decentralized multiplayer bandits is almost equivalent to the centralized one for the considered model. However, this equivalence may not hold without sensing. In that case, we presented an algorithm with a logarithmic regret, assuming a lower bound of the mean rewards is known. This adds a factor M to the regret achieved with sensing.

Our major claim is that synchronization should not be considered anymore, unless communication is allowed. We thus proposed the first algorithm for the dynamic model without sensing reaching a logarithmic regret. It can even be extended to the case of user-dependent rewards.

Decentralized multiplayer bandits should thus be considered either while allowing limited communication among players or in the dynamic model. As an example, the arrival or leaving of a player could be communicated to other players. Also, classical algorithms as UCB need a deeper investigation, as they tend to perform well in practice, although they significantly fail at times.

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A Complementary material for Section 2

A.1 Algorithm description

We here describe in detail SIC-MMAB algorithm. All the pseudocodes are described from the point of view of a single player, which corresponds to the natural way to describe a decentralized algorithm. First, this algorithm relies on the Musical Chairs algorithm, introduced by Rosenski et al. [2016]. We recall it in Pseudocode 3.

MusicalChairs Protocol

Input: $[K_p]$ (set of active arms), T_0 (time of procedure)

for T_0 time steps **do**
  if not fixed yet **then**
    Sample k chosen uniformly at random in $[K_p]$
    if $\eta_k(t) = 0$ ($r_k(t) > 0$ for No Sensing setting) **then**
      `Fixed` $\leftarrow k$ **end if** *# The player stays in k if no collision*
  else Sample `Fixed` **end if**
end for
return `Fixed` (or leave the game if not fixed) *# External rank*

Pseudocode 3: Reach an orthogonal setting in T_0 steps

The initialization phase then consists in a second procedure. Its purpose is to estimate M and to assign different ranks in $[M]$ to all players. This procedure is illustrated in Figure 3 and described in Pseudocode 4 below. SIC-MMAB is finally described in Algorithm 1.

Estimate_M Protocol

Input: $k \in [K]$ (external rank)

Initialize $M \leftarrow 1$, $j \leftarrow 1$ and $\pi \leftarrow k$ *# estimates of M and the internal rank*
for $2k - 1$ time steps **do**
  Pull π; **if** $\eta_\pi(t) = 1$ **then** $M \leftarrow M + 1$ and $j \leftarrow j + 1$ **end if** *# increases if collision*
end for
for $2(K - k) + 1$ time steps **do**
  $\pi \leftarrow \pi + 1 \pmod K$ and pull π *# sequential hopping*
  if $\eta_\pi(t) = 1$ **then** $M \leftarrow M + 1$ **end if** *# increases if collision*
end for
return M, j

Pseudocode 4: Estimate M and assign ranks among the players

A.2 Proofs

A.2.1 Lemma 1

Proof. We show that for a single player j, her probability to encounter only collisions until time T_0 is lower than $\exp\left(-\frac{T_0}{K}\right)$. The union bound over the M players then yields the desired result.

The probability of having no collision at time t for a single player is lower bounded by $\frac{1}{K}$, since there is at least one arm that is not played by all the other players at each time step. It thus holds:

$$\mathbb{P}\left[\forall t \leq T_0, \eta^j(t) = 1\right] \leq \left(1 - \frac{1}{K}\right)^{T_0} \leq \exp\left(-\frac{T_0}{K}\right)$$

□

A.2.2 Proposition 1

Proof. With probability at least $1 - M\exp\left(-\frac{T_0}{K}\right)$, the initialization is successful, i.e., all players have been assigned different ranks. The remaining of the proof is conditioned to that event.

Hoeffding inequality gives the following, classical lemma in MAB.

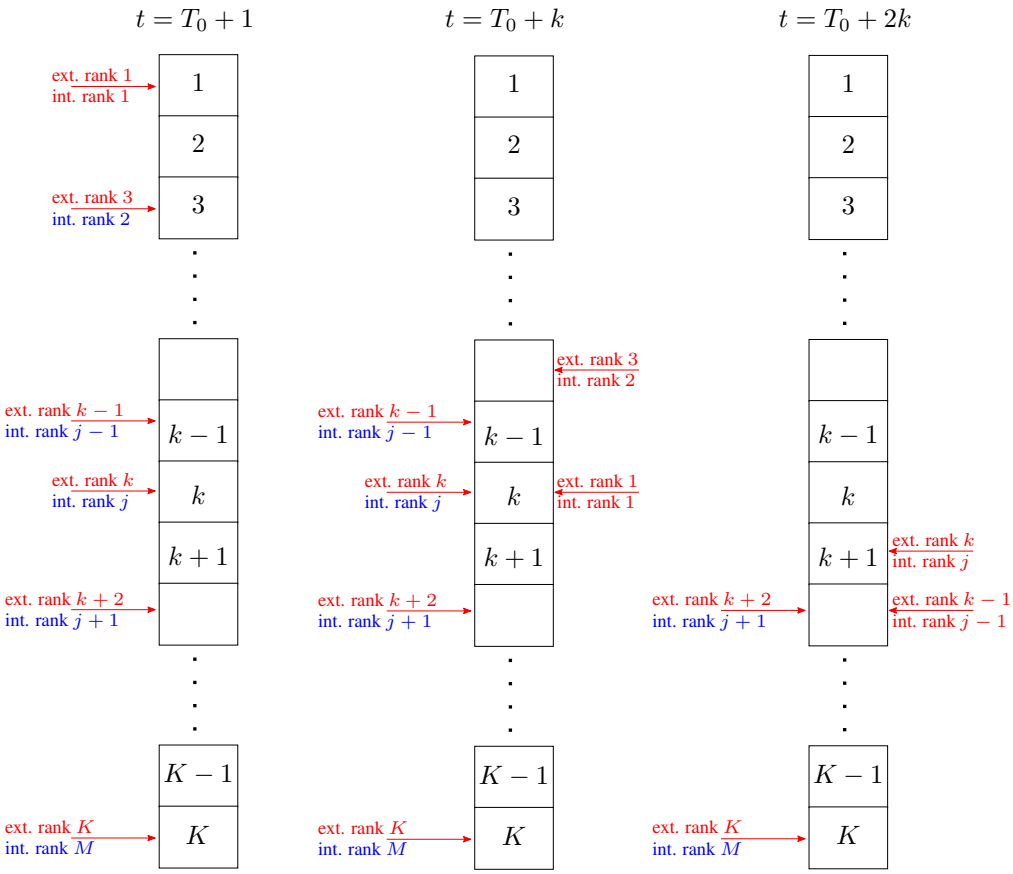

Figure 3: Estimating M and the internal rank in a time $2K$. A column represents the set of arms. The left (resp. right) arrows represent the choice of a player fixed to an arm (resp. sequentially hopping). The information written in red (resp. blue) is known (resp. unknown) to the player at the considered time step.

Lemma 4. $\mathbb{P}[\exists p \leq n : |\hat{\mu}_k(p) - \mu_k| \geq B_{T_k(p)}] \leq \frac{2n}{T}$

We first consider an optimal arm k. Let $\Delta_k = \mu_k - \mu_{(M+1)}$ be the gap between the arm k and the first suboptimal arm. We assume $\Delta_k > 0$ here, the case of equality holds considering $\frac{\log(T)}{0} = \infty$. Let s_k be the first integer such that $4B_{s_k} \leq \Delta_k$.

If we denote $T(p) = \sum_{l=1}^{p} M_l 2^l$ the number of times an active arm has been pulled after the p-th exploration phase, it holds that

$$2^{p+1} \leq T(p+1) \leq 3T(p) \qquad \text{as } M_p \text{ is non-increasing.} \tag{2}$$

Thus, for some $p \in \mathbb{N}$, $T(p-1) < s_k \leq T(p)$. The triangle inequality for such a p, for any active suboptimal arm i, yields with the probability considered in Lemma 4: $\hat{\mu}_k(p) - B_{T_k(p)} \geq \hat{\mu}_i(p) + B_{T_i(p)}$

Algorithm 1 SIC-MMAB algorithm

Input: T (horizon), T_0 (time of Musical Chairs), $B_s = \sqrt{\frac{2\log(T)}{s}}$ (error bound)

Initialization Phase:
$k \leftarrow$ MusicalChairs $([K], T_0)$
$(M_p, j) \leftarrow$ Estimate_M (k) *# estimated number of players and assigned internal rank*
Initialize $p \leftarrow 1$; $[K_p] \leftarrow [K]$ and $\mathbf{S}, \mathbf{T} \leftarrow$ Zeros(K) *# Zeros(K) returns a vector of length K containing only zeros*
while not fixed yet **do**
  Exploration Phase:
  $\pi \leftarrow j$ and $\mathbf{S_r} \leftarrow$ Zeros(K) *# start of a new phase*
  for $K_p 2^p$ time steps **do**
    $\pi \leftarrow \pi + 1 \pmod{[K_p]}$ and pull π *# sequential hopping*
    $S[\pi] \leftarrow S[\pi] + r_\pi(t)$; $S_r[\pi] \leftarrow S_r[\pi] + r_\pi(t)$ and $T[\pi] \leftarrow T[\pi] + 1$
  end for
  Communication Phase:
  Define $E_p := \{(i, l, k) \in [M_p] \times [M_p] \times [K_p] \mid i \neq l\}$
  for $(i, l, k) \in E_p$ **do** *# Player i sends stats about arm k to player l*
    if $i = j$ **then** Send $(l, S_r[k], p, j, [K_p])$ *# player communicating*
    else if $l = j$ **then** *# player receiving*
      $s \leftarrow$ Receive$(p, j, [K_p])$; $S[k] \leftarrow S[k] + s$ and $T[k] \leftarrow T[k] + 2^p$
    else for $p + 1$ time steps **do** Pull j **end for** *# wait while others communicate*
    end if
  end for
  Update Statistics:
  Rej $\leftarrow$ set of active arms k verifying $\#\{i \in [K_p] \mid \frac{S[i]}{T[i]} - B_{T[i]} \geq \frac{S[k]}{T[k]} + B_{T[k]}\} \geq M_p$
  Acc $\leftarrow$ set of active arms k verifying $\#\{i \in [K_p] \mid \frac{S[k]}{T[k]} - B_{T[k]} \geq \frac{S[i]}{T[i]} + B_{T[i]}\} \geq K_p - M_p$, ordered according to their indices
  if $j \leq$ `length`(Acc) **then** `Fixed` $\leftarrow$ Acc$[j]$ *# Start exploiting Acc$[j]$*
  else *# Update all the statistics*
    $M_p \leftarrow M_p -$ `length`(Acc); $j \leftarrow j -$ `length`(Acc) and $[K_p] \leftarrow [K_p] \setminus$ (Acc $\cup$ Rej)
  end if
  $p \leftarrow p + 1$
end while
Exploitation Phase: Pull `Fixed` until T

So the arm k is accepted after at most p phases. Using the same argument as in [Perchet et al., 2015], it holds $s_k = \mathcal{O}\left(\frac{\log(T)}{(\mu_k - \mu_{(M+1)})^2}\right)$, and also for $T_k(p)$ thanks to Equation (2). Also, a sub-optimal arm k can not be wrongly accepted (or inversely for optimal rejected arms) conditionally to the same event, as it can not dominate any optimal arm in term of confidence intervals.

The proof for the sub-optimal case is similar, if we denote $\Delta_k = \mu_{(M)} - \mu_k$.

Lemma 4 gives that all of this happens with probability $1 - \mathcal{O}\left(\frac{KN}{T}\right)$, where N is the number of exploration phases. By construction of the algorithm, $N \leq \lceil \log_2(T) \rceil$, so overall, with probability

$1 - \mathcal{O}\left(\frac{K\log(T)}{T}\right)$. $\square$

In the following, we keep the notations $t_k = \min\left\{\frac{c\log(T)}{\left(\mu_k - \mu_{(M)}\right)^2},\ T\right\}$, where c is a universal constant such that with the probability considered in Proposition 1, the number of exploration pulls before accepting/rejecting k is at most t_k.

A.2.3 Lemma 2

Proof. We (re)define the following: $\hat{t}_j$ the number of exploratory pulls before j is accepted/rejected, $T(p) = \sum_{l=1}^{p} 2^l M_l$ and N the total number of exploration phases.

$T(p)$ describes the total number of exploration pulls processed at the end of the p-th exploration phase on every active arm for $p < N$. Since the N-th phase may remain uncompleted, $T(N)$ is then greater that the number of exploration pulls at the end of the N-th phase.

With probability $1 - \mathcal{O}\left(\frac{K\log(T)}{T} + M\exp\left(-\frac{T_0}{K}\right)\right)$, the initialization is successful, any arm is correctly accepted or rejected and $\hat{t}_k \leq t_k$ for all k. The remaining of the proof is conditioned to that event.

Let us consider the optimal arm k. During a phase p, there are two possibilities:

- either k has already been accepted, i.e., $\hat{t}_k \leq T(p-1)$. Then the arm k is pulled the whole phase, i.e., $K_p 2^p$ times.
- or k is still active. Then it is pulled 2^p times by each active player, i.e., it is pulled $M_p 2^p$ times in total.

As $t_k \geq \hat{t}_k$, it then holds that $T_k^{\text{explo}} \geq T^{\text{explo}} - \sum_{p=1}^{N} 2^p (K_p - M_p)\mathbb{1}_{t_k > T(p-1)}$.

Notice that $K_p - M_p$ is the number of active suboptimal arms. As $t_k \geq \hat{t}_k$, it also holds that $K_p - M_p \leq \sum_{j>M} \mathbb{1}_{t_{(j)} > T(p-1)}$. We thus get that

$$T_k^{\text{explo}} \geq T^{\text{explo}} - \sum_{j>M}\sum_{p=1}^{N} 2^p \mathbb{1}_{\min(t_{(j)}, t_k) > T(p-1)} \tag{3}$$

The double sum actually corresponds to the number of times a suboptimal arm is pulled instead of k. The regret due to the optimal arms is then given by summing Equation (3) over all optimal arms.

$$\sum_{k\leq M} (\mu_{(k)} - \mu_{(M)})(T^{\text{explo}} - T_{(k)}^{\text{explo}}) \leq \sum_{j>M}\sum_{k\leq M}\sum_{p=1}^{N} 2^p (\mu_{(k)} - \mu_{(M)})\mathbb{1}_{\min(t_{(j)}, t_{(k)}) > T(p-1)} \tag{4}$$

Let us define $A_j = \sum_{k\leq M}\sum_{p=0}^{N} 2^p (\mu_{(k)} - \mu_{(M)})\mathbb{1}_{\min(t_j, t_{(k)}) > T(p-1)}$ the cost associated to the suboptimal arm j. The regret in Equation (4) then corresponds to the sum of all $A_{(j)}$ for $j > M$.

Recall that $t_{(k)} = \min\left(\frac{c\log(T)}{(\mu_{(k)}-\mu_{(M)})^2},\ T\right)$ for a universal constant c. Thus, if we define $\Delta(p) = \sqrt{\frac{c\log(T)}{T(p-1)}}$, the inequality $t_{(k)} > T(p-1)$ becomes $\mu_{(k)} - \mu_{(M)} < \Delta(p)$. We also write N^j the last phase with arm j still active. It follows:

$$
\begin{aligned}
A_j &\leq \sum_{k\leq M}\sum_{p=1}^{N^j} 2^p\Delta(p)\mathbb{1}_{t(k)>T(p-1)} \\
&\leq \sum_{p=1}^{N^j}\Delta(p)\left(T(p)-T(p-1)\right) = c\log(T)\sum_{p=1}^{N^j}\Delta(p)\left(\frac{1}{\Delta(p+1)}+\frac{1}{\Delta(p)}\right)\left(\frac{1}{\Delta(p+1)}-\frac{1}{\Delta(p)}\right) \\
&\leq (1+\sqrt{3})c\log(T)\sum_{p=1}^{N^j}\left(\frac{1}{\Delta(p+1)}-\frac{1}{\Delta(p)}\right) && \text{thanks to Equation (2)} \\
&\leq (1+\sqrt{3})c\log(T)\frac{1}{\Delta(N^j+1)} && \text{by convention, } \frac{1}{\Delta(1)} = 0
\end{aligned}
$$

By definition of N^j, we have $t_j \geq T(N^j)$. Thus, $\Delta(N^j+1) \geq \sqrt{\frac{c\log(T)}{t_j}}$, thus $A_j \leq (1+\sqrt{3})\sqrt{c\,t_j\log(T)}$. The result follows since $t_j = \mathcal{O}\left(\min\left\{\frac{\log(T)}{(\mu_{(M)}-\mu_j)^2}, T\right\}\right)$. □

B Complementary material for Section 3

B.1 Algorithm description

SIC-MMAB2 algorithm is described in this section. We use the same notations for M_p and K_p.

B.1.1 Initialization phase

The objective of the initialization phase is to estimate M and is explained in Pseudocode 5. The idea is the same as for SIC-MMAB, but as already mentioned, instead of a single time step, a number of $\frac{\log(T)}{\mu_{\min}}$ time steps is needed to correctly transmit a bit with high probability[2].

Estimate_M_NoSensing Protocol

Input: k (external rank), T_c (time)

Initialize $M \leftarrow 1$ and $\pi \leftarrow k$
for $n = 1, \ldots, 2K$ **do**
  Initialize $r \leftarrow 0$
  if $n \geq 2k$ **then** $\pi \leftarrow \pi + 1 \pmod K$ **end if** *# sequential hopping*
  for T_c time steps **do** Pull π and update $r \leftarrow r + r_\pi(t)$ **end for**
  if $r = 0$ **then** $M \leftarrow M + 1$ **end if** *# increases if collision*
end for
return M

Pseudocode 5: Estimate M for the No Sensing model

[2] As mentioned in Section 3, $T_c = \lceil\frac{\log(T)}{\mu_{\min}}\rceil$.

B.1.2 Exploration phases

Each exploration phase is split into two parts. During the first one, each player fixes to an arm following Musical Chairs procedure. At its end, players are in an orthogonal setting and can thus start the second part, where they hop sequentially and explore the active arms without any collision. The decisions for accepting/rejecting arms are still based on the exploration pulls as for SIC-MMAB. The differences with the exploration of SIC-MMAB are the following:

1. Statistics are not shared among players; this induces an additional M factor in the regret
2. A Musical Chairs procedure is added at the beginning of a new exploration phase, if there was at least one declaration or fixation block in the previous communication phase. This procedure is needed for players to reach an orthogonal setting before the sequential hopping, as internal ranks are not updated in SIC-MMAB2. Figures 4 and 5 below illustrate when such a procedure is added. It corresponds to lines 8-11 in Algorithm 2.

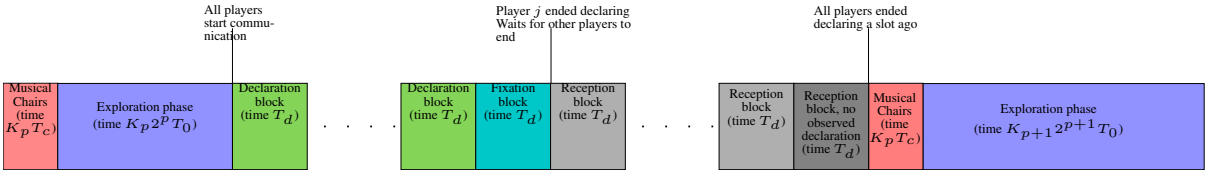

Figure 4: Alternating between fixation, exploration and declaration blocks. Case where the players declare sub-optimal and try to occupy (in vain) optimal arms

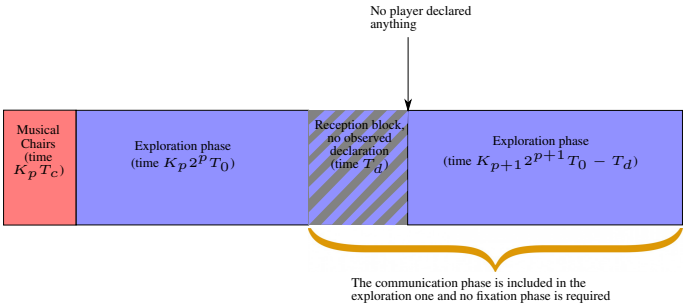

Figure 5: Alternating between fixation, exploration and declaration blocks. Case with no declaration. In that case, the single declaration block, which just consists in sequential hopping, is included in the next exploration phase (lines 10-11 in Algorithm 2). No fixation phase is needed in that case.

B.1.3 Communication phase

Notice that all players are in a communication phase at the same time. However, this phase is decomposed into blocks of same length T_d (to keep synchronization). A block can be of three different types, and the type of a block does not need to be the same for all players, as illustrated by Figure 4. Types are the following:

Declaration block for player j: she communicates to other players that an arm is sub-optimal.

Fixation block for player j: she tries to occupy any arm that she detected as optimal. If she succeeds, she exploits that arm until the end.

Reception block for player j: she just hops sequentially, in order to detect other players' declarations.

Player j starts the communication phase with declaration blocks, one per arm detected as sub-optimal [3]. She then proceeds to a fixation block, had she detected any arm as optimal during the last exploitation phase. She then proceeds to reception blocks until no new declaration is detected. As soon as no new declaration is detected, she starts a new exploration phase.

Notice that players keep receiving declarations from other players in any type of block.

Declaration block: In a declaration block, player j follows **Declare Protocol**, described in Pseudocode 6. The idea is to frequently sample the sub-optimal arm to send a "signal" to the other players. They will detect this signal by observing a significant loss in the empirical reward of this arm. However, a player sending a signal should also be able to detect signals on other arms sent by other players. That is the reason why in order to declare an arm as sub-optimal, a player randomly chooses between pulling this arm and sequentially hopping.

Declare Protocol

Input: k (arm to declare), T_d (time of block), π (first arm to pull in sequential hopping), $\mathbf{S}$ and $\mathbf{T}$ (exploration statistics), $[K_p]$ (active arms)

Initialize $\mathbf{s}, \mathbf{t} \leftarrow \text{Zeros}(K)$
for T_d time steps **do**
  Pull arm $i = \begin{cases} k \text{ with probability } \frac{1}{2} \\ \pi \text{ with probability } \frac{1}{2} \end{cases}$
  $s[i] \leftarrow s[i] + r_i(t)$; $t[i] \leftarrow t[i] + 1$ and $\pi \leftarrow \pi + 1 \pmod{[K_p]}$
end for
$d \leftarrow$ set of active arms i verifying $|\frac{S[i]}{T[i]} - \frac{s[i]}{t[i]}| \geq \frac{S[i]}{4T[i]}$
return $d \cup \{k\}$ *# arms declared during the block*

Pseudocode 6: Declare arm k as suboptimal

Lemma 6 gives an appropriate choice for T_d such that the declaration is detected by every player, without detecting any false positive declaration, no matter the block they are currently proceeding, with high probability.

To give some intuition, let $\hat{\mu}_i$ and $\hat{r}_i$ be respectively the empirical reward during the exploration phases and during the last communication block for arm i and player j. Arm i is understood as declared, i.e., another player is detected as sending a signal or exploiting this arm if:

$$|\hat{\mu}_i - \hat{r}_i| \geq \frac{\hat{\mu}_i}{4} \tag{5}$$

Lemma 6 states that players will only detect arms declared as sub-optimal or exploited by a player with high probability. However, using the last reception block where there is no new declaration, it is easy to distinguish optimal arms exploited by players from the sub-optimal ones. Indeed,

[3] Of course, she does not declare any arm previously declared by another player in the same phase.

for the optimal ones, only 0 are observed during this last phase, while for the sub-optimal ones, at least one positive reward will be observed during the phase of length T_d with probability at least $1 - \frac{1}{T}$, since no player pulls it except for the sequential hopping.

Fixation block: In a fixation block, player j proceeds to **Occupy Protocol**, described in Pseudocode 7. She continues sequential hopping on the active arms and decides to occupy an arm as soon as any optimal arm returns a reward 1 (i.e., without collision at this step). In that case, she exploits this arm until the final horizon T. At the end of a block, if she did not manage to occupy any detected optimal arm, then all of them are occupied by other players with high probability. Declarations of other players are detected following the rule of Equation (5).

Occupy Protocol

Input: A (set to occupy), T_d (time of block), π (first arm to pull), $\mathbf{S}$ and $\mathbf{T}$ (exploration statistics), $[K_p]$ (active arms)

Initialize $\mathbf{s}, \mathbf{t} \leftarrow \text{Zeros}(K)$
for T_d time steps **do**
  if not fixed yet **then**
    Pull π
    if $\pi \in A$ and $r_\pi(t) > 0$ **then** `Fixed` $\leftarrow \pi$ **end if** *# no collision on optimal arm*
    $s[\pi] \leftarrow s[\pi] + r_\pi(t)$; $t[\pi] \leftarrow t[\pi] + 1$ and $\pi \leftarrow \pi + 1 \pmod{[K_p]}$
  else Pull `Fixed` **end if**
end for
$d \leftarrow$ set of active arms k verifying $|\frac{S[k]}{T[k]} - \frac{s[k]}{t[k]}| \geq \frac{S[k]}{4T[k]}$
return (`Fixed`, d)

Pseudocode 7: Try to start exploiting an arm among A

Reception block: In a reception block, player j simply sequentially hops and detects the declarations of other players, following the rule of Equation (5). This corresponds to **Receive Protocol**, described in Pseudocode 8.

Receive Protocol

Input: T_d (time of block), π (first arm to pull), $\mathbf{S}$ and $\mathbf{T}$ (exploration statistics), $[K_p]$ (active arms)

Initialize $\mathbf{s}, \mathbf{t} \leftarrow \text{Zeros}(K)$
for T_d time steps **do**
  Pull π
  Update $s[\pi] \leftarrow s[\pi] + r_\pi(t)$; $t[\pi] \leftarrow t[\pi] + 1$ and $\pi \leftarrow \pi + 1 \pmod{[K_p]}$
end for
$d \leftarrow$ set of active arms k verifying $|\frac{S[k]}{T[k]} - \frac{s[k]}{t[k]}| \geq \frac{S[k]}{4T[k]}$
return $(d, \mathbf{s}, \mathbf{t})$

Pseudocode 8: Detect other players' declarations (and wait)

Notice that every active player will at least proceed to one reception block per communication phase (if she does not end occupying an optimal arm). The last reception block is considered as the block where no new declaration is detected. This block is thus the same for all active players.

Moreover, the arms giving 0 reward during this last reception block correspond to optimal arms occupied by other players. This allows to distinguish between arms occupied (that are optimal), and just declared by some player (that are sub-optimal). This distinction is described in Pseudocode 9. As a consequence, active players share a common set of active arms $[K_p]$ and number of active players M_p at the end of each communication phase.

Update

Input: Decl (declared arms), s (statistics of last reception block)

Opt $\leftarrow \{i \in \text{Decl} \mid s[i] = 0\}$

Subopt $\leftarrow$ Decl $\setminus$ Opt

return (Opt, Subopt)

Pseudocode 9: Distinguish the declared suboptimal and optimal arms after a communication phase

The complete description of SIC-MMAB2 is given in Algorithm 2 below.

B.2 Regret Analysis

This section is devoted to the proof of Theorem 2. It first proves several required lemmas.

The same decomposition of the regret is used for SIC-MMAB2: $R_T = R^{\text{init}} + R^{\text{explo}} + R^{\text{comm}}$, but in this section, a communication step is defined as a time step in a communication phase where there is at least a player using **Declare** or **Occupy protocol**, and $T_{\text{init}} = 3KT_c$. Notice that the last reception block of a communication phase then counts as communication steps only if there were declarations in previous blocks of the communication phase.

B.2.1 Initialization Regret

The initialization phase lasts $3KT_c$ steps. Lemma 5 claims that the initialization is successful, meaning all players perfectly know M after this phase, with a probability depending on T_c.

Lemma 5. *With probability $1 - \mathcal{O}\left(MK \exp\left(-\mu_{\min} T_c\right)\right)$, at the end of this procedure, every player has a correct estimation of M and players are in an orthogonal setting.*

Proof. Similarly to the proof of Lemma 1, the probability to encounter a positive reward for a player during the Musical Chairs procedure at time step t is lower bounded by $\frac{\mu_{\min}}{K}$. Hence using the same arguments, with probability $1 - \mathcal{O}\left(M \exp\left(-\mu_{\min} T_c\right)\right)$, all the players are pulling different arms after a time KT_c.

We now consider the **Estimate_M_NoSensing protocol**. Every time a player sends a bit to another player, it will be detected. Let us now bound the probability that a player detects a "collision" with another player while there is not. This is the case when she encounters T_c successive zero rewards on an arm while she is the only player to sample it. This happens with probability smaller than $\exp\left(-\mu_{\min} T_c\right)$ for a single player for a single block of T_c steps. The union bound over the M players and the $2K$ blocks of T_c steps yields the results. □

B.2.2 Communication regret

Lemma 6 provides the properties and regret of the algorithm during the communication phases.

Lemma 6. *Let the length of a block be such that $T_d = \lceil \frac{2400 K_p \log(T)}{\mu_{\min}} \rceil$, then, conditionally to the outcome of the previous Musical Chairs procedures:*

Algorithm 2 SIC-MMAB2 algorithm

Input: T (horizon), T_c (time), $\mu_{\min}$ (lower bound of means), $B_s = \sqrt{\frac{2\log(T)}{s}}$ (error bound)

1: **Initialization Phase:**
2: $k \leftarrow$ MusicalChairs $([K], KT_c)$
3: $M_p \leftarrow$ Estimate_M_NoSensing (k, T_c)
4: Initialize $T_0 \leftarrow \lceil \frac{2400\log(T)}{\mu_{\min}} \rceil$; Decl $\leftarrow \{0\}$; $[K_p] \leftarrow [K]$ and $\mathbf{S}, \mathbf{T} \leftarrow$ Zeros(K)
5: **for** $p = 1, \ldots, \infty$ **do**
6: **Exploration Phase:**
7: $T_{\text{expl}} \leftarrow K_p 2^p T_0$ and $n_0 \leftarrow 0$
8: **if** `length`(Decl)> 0 **then** *# there was a declaration in the previous phase*
9: $\pi \leftarrow$ MusicalChairs $([K_p], K_p T_c)$
10: **else** $\mathbf{S} \leftarrow \mathbf{S} + \mathbf{s}$; $\mathbf{T} \leftarrow \mathbf{T} + \mathbf{t}$ and $n_0 \leftarrow T_d$ *# statistics of the last reception block*
11: **end if**
12: **for** $T_{\text{expl}} - n_0$ time steps **do** *# start exploration*
13: Pull π; $S[\pi] \leftarrow S[\pi] + r_\pi(t)$; $T[\pi] \leftarrow T[\pi] + 1$ and $\pi \leftarrow \pi + 1 \pmod{[K_p]}$
14: **end for**
15: **Communication Phase:**
16: Initialize $T_d \leftarrow K_p T_0$ and Decl as empty set
17: Rej $\leftarrow$ set of active arms k verifying $\#\{i \in [K_p] \mid \frac{S[i]}{T[i]} - B_{T[i]} \geq \frac{S[k]}{T[k]} + B_{T[k]}\} \geq M_p$
18: Acc $\leftarrow$ set of active arms k verifying $\#\{i \in [K_p] \mid \frac{S[k]}{T[k]} - B_{T[k]} \geq \frac{S[i]}{T[i]} + B_{T[i]}\} \geq K_p - M_p$
19: **while** Rej $\setminus$ Decl $\neq \emptyset$ **do** *# declaration blocks*
20: Let $k \in$ Rej $\setminus$ Decl
21: $d \leftarrow$ Declare$(k, T_d, \pi, \mathbf{S}, \mathbf{T}, [K_p])$ and add d to Decl
22: **end while**
23: **if** Acc $\setminus$ Decl $\neq \emptyset$ **then** *# fixation block*
24: $(\texttt{Fixed}, d) \leftarrow$ Occupy(Acc $\setminus$ Decl$, T_d, \pi, \mathbf{S}, \mathbf{T}, [K_p])$ and add d to Decl
25: **if** successfully fixed **then** go to line 35 (break) **end if**
26: **end if**
27: Reset $d \leftarrow \{0\}$
28: **while** $d \neq \emptyset$ **do** *# reception blocks*
29: $(d, \mathbf{s}, \mathbf{t}) \leftarrow$ Receive$(T_d, \pi, \mathbf{S}, \mathbf{T}, [K_p])$ and add d to Decl *# $\mathbf{s}$ and $\mathbf{t}$ are the statistics*
30: **end while**
31: **Update Statistics:**
32: (Opt, Subopt) $\leftarrow$ Update(Decl, s)
33: $[K_p] \leftarrow [K_p] \setminus$ (Opt $\cup$ Subopt) and $M_p \leftarrow M_p -$ `length`(Opt)
34: **end for**
35: **Exploitation Phase:** Pull `Fixed` until T

1. *With probability $1 - \mathcal{O}\left(\frac{M}{T}\right)$, a player j declaring an arm i as sub-optimal will be successfully detected by all other players;*

2. *With probability $1 - \mathcal{O}\left(\frac{MK}{T}\right)$, no player will detect a false declaration during the declaration block (i.e., no arm is detected as sub-optimal if there was no declaration or if it is not occupied*

by another player);

3. *With probability* $1-\mathcal{O}\left(\frac{M}{T}\right)$*, if player* j *starts occupying arm* k*, it is detected as a declaration by active players (following the rule of Equation* (5)*).*

Thus, with probability $1-\mathcal{O}\left(MK\exp(-\mu_{\min}T_c)+(K+\log(T))\frac{KM}{T}\right)$*, all communication phases are successful, i.e., all declarations are correctly detected and no false declarations are detected. Then*

$$R^{\text{comm}} = \mathcal{O}\left(\frac{MK^2}{\mu_{\min}}\log(T)\right).$$

Proof. We first prove the three points conditionally to the successes of the previous Musical Chairs procedures.

1) We prove this point in the more general case where the declaration of an arm follows the sampling: $\begin{cases}\text{Pull } k \text{ with probability } \lambda_d \\ \text{Sample according to the sequential hopping on } [K_p], \text{ otherwise}\end{cases}$

First, denote by $T_{i'}^{j'}$ the number of pulls by player j' on arm i' during a block of length T_d. Using the Chernoff bound, it holds:

$$\begin{aligned}\mathbb{P}\left[T_{i'}^{j'} \leq \frac{(1-\lambda_d)T_d}{2K_p}\right] &\leq \exp\left(-\frac{(1-\lambda_d)T_d}{8K_p}\right) \\ &\leq \frac{1}{T} \qquad \text{as long as } \frac{(1-\lambda_d)T_d}{8K_p} \geq \log(T)\end{aligned} \tag{6}$$

This last condition holds with T_d chosen as in Equation (7) and this inequality holds, no matter the type of block player j' is proceeding.

With probability $1-\mathcal{O}\left(\frac{K_pM_p}{T}\right)$, all the T_i^j are thus greater than $\frac{(1-\lambda_d)T_d}{2K_p}$. Let r_i and $\hat{r}_i$ respectively denote the expected and the empirical observed rewards of the arm i during this declaration phase for player j. Assume that the arm i is declared as sub-optimal by another player during the considered phase. It then holds that $r_i \leq \mu_i(1-\lambda_d)$.

Chernoff bound provides the following inequalities:

$$\mathbb{P}\left[|\hat{r}_i - r_i| \geq \frac{\lambda_d\mu_i}{5}\delta_{T_i^j}\right] \leq \frac{2}{T} \qquad \text{with } \delta_s = \max\left\{5\sqrt{\frac{6\log(T)}{s\lambda_d^2\mu_{\min}}}, \frac{30\log(T)}{s\lambda_d\mu_{\min}}\right\}$$

$$\text{and } \mathbb{P}\left[|\hat{\mu}_i - \mu_i| \geq \frac{\lambda_d\mu_i}{5}\delta_{T_i^{j,\,\text{explo}}}\right] \leq \frac{2}{T} \qquad \text{for the exploration phases}$$

$$\text{With the specific choice of} \qquad T_d \geq \frac{300K_p\log(T)}{(1-\lambda_d)\lambda_d^2\mu_{\min}} \tag{7}$$

it holds that $\delta_{T_i^j} \leq 1$ with probability at least $1-\frac{1}{T}$ from Equation (6). This also holds for the exploration pulls, since the first exploration phase lasts at least T_d steps.

We then consider the high probability event $|\hat{r}_i - r_i| \leq \frac{\lambda_d}{5}\mu_i$ and $|\hat{\mu}_i - \mu_i| \leq \frac{\lambda_d}{5}\mu_i$.

As $\lambda_d \leq 1$, the second inequality yields: $\frac{5}{6}\hat{\mu}_i \leq \mu_i \leq \frac{5}{4}\hat{\mu}_i$

If i is declared by a player, $\mu_i - r_i \geq \lambda_d \mu_i$, then:

$$|\hat{\mu}_i - \hat{r}_i| \geq \frac{3\lambda_d}{5}\mu_i \geq \frac{\lambda_d}{2}\hat{\mu}_i \tag{8}$$

This means that with the detection rule described in Appendix B.1.3 for $\lambda_d = \frac{1}{2}$, for a single arm i and player j, with probability at least $1 - \frac{2}{T}$, the player will correctly detect the declaration of arm i as sub-optimal by (at least) another player.

2) As in the first point, with probability greater than $1 - \mathcal{O}\left(\frac{1}{T}\right)$, it holds $|\hat{r}_i - r_i| \leq \frac{\lambda_d}{5}\mu_i$. The case of neither declaration nor exploitation by any other player actually corresponds to $r_i = \mu_i$. Thus with probability greater than $1 - \mathcal{O}\left(\frac{1}{T}\right)$, we can rewrite Equation (8) of the first case into:

$$|\hat{\mu}_i - \hat{r}_i| \leq \frac{2\lambda_d}{5}\mu_i \leq \frac{\lambda_d}{2}\hat{\mu}_i$$

Considering all arms and players yields the second point.

3) The same argument as in Lemma 5 gives that with probability $1-\mathcal{O}\left(\frac{1}{T}\right)$, player j will actually starts occupying k after a time $t_{\text{fix}} \leq \frac{K_p \log(T)}{\mu_{\min}} \leq \frac{T_d}{2400}$.

Chernoff bound then provides a bound on the total reward $X_k^{j'}$ observed by j' on k for a time t_{fix}, i.e., for $T_k^{j'} \leq \frac{\log(T)}{\mu_{\min}}$ pulls on k.

$$\mathbb{P}\left[X_k^{j'} \geq 4\mu_k \frac{\log(T)}{\mu_{\min}}\right] \leq \exp\left(-\frac{3\mu_k \log(T)}{3\mu_{\min}}\right) \leq \frac{1}{T} \tag{9}$$

Thus, Equation (9) claims that with probability $1 - \mathcal{O}\left(\frac{1}{T}\right)$, $X_k^{j'} \leq \frac{4\mu_k \log(T)}{\mu_{\min}}$.

However, k will be occupied after that point and no other positive reward will be observed by player j'. As a consequence, her empirical reward on k will be for this block $\hat{r}_k^{j'} \leq \frac{2X_k^{j'} K_p}{(1-\lambda_d)T_d} \leq \frac{\mu_k}{75}$. Using the same argument as in points 1) and 2), this guarantees $|\hat{\mu}_k^{j'} - \hat{r}_k^{j'}| \geq \frac{\hat{\mu}_k^{j'}}{4}$ and the result follows.

Conditionally to the successes of all previous Musical Chairs procedures (i.e., players end these procedures in orthogonal settings), these three points imply that, with probability $1 - \mathcal{O}\left(\frac{MK}{T}\right)$, the communication block will be successful: all declarations are correctly detected, all detected optimal arms are exploited by a single player and there is no false detection of declaration.

Let N be the total number of exploration phases. By construction of the algorithm, $N \leq \lceil \log_2(T) \rceil$. Also there can not be two different blocks used to declare or occupy the same arm, conditioned to the successes of the previous communication blocks and Musical Chairs. Hence, conditionally to this event, there will be at most $N+K$ communication blocks, each succeeding with probability $1 - \mathcal{O}\left(\frac{KM}{T}\right)$ and there will be at most K Musical Chairs procedures, each also succeeding with probability $1-\mathcal{O}(M\exp(-\mu_{\min}T_c))$. Using a chain rule argument, all the communication protocols and Musical Chairs procedures are successful with probability $1 - \mathcal{O}\Big(KM\exp(-\mu_{\min}T_c) + (K+N)\frac{KM}{T}\Big)$.

Thus the length of Comm is at most $2KT_d$, since only the communication phases with at least a Declaration or Fixation block are counted. This leads to Lemma 6. □

B.2.3 Exploration regret

Conditionally to the successes of the initialization phase, all the communication phases and all the Musical Chairs procedures at the beginning of exploration phases, the exploration (except the Musical Chairs) will be collision-less. Using the arguments of Lemma 2, we provide an upper bound for the exploration regret of SIC-MMAB2.

Lemma 7. *With probability* $1 - \mathcal{O}\left(KM\exp\left(-\mu_{\min}T_c\right) + (K + \log(T))\frac{KM}{T}\right)$*:*

$$R^{\text{explo}} \leq \mathcal{O}\left(\sum_{k>M} \min\left\{\frac{M\log(T)}{\mu_{(M)} - \mu_{(k)}}, \sqrt{T\log(T)}\right\} + \frac{MK^2}{\mu_{\min}}\log(T)\right)$$

Proof. First, as already claimed in the proof of the communication regret, the initialization, all the communication blocks and Musical chairs procedures succeed and there are at most K Musical Chairs procedures during the exploration, with probability $1 - \mathcal{O}\Big(MK\exp(-\mu_{\min}T_c) + (K + \log(T))\frac{KM}{T}\Big)$. The remaining of the proof is conditioned to this event. A single Musical Chairs procedure lasts a time $\frac{K_p\log(T)}{\mu_{\min}}$, hence the total cost of the Musical Chairs is smaller than $\frac{MK^2\log(T)}{\mu_{\min}}$.

We now consider the cost of exploration pulls (except for the Musical Chairs). We denote N the number of exploration phases that will be run in the algorithm and the same notation as in Appendix A.2.2 concerning Δ_k.

Since the players do not share their statistics, it can be shown with the same arguments as in Appendix A.2.2 that a sub-optimal arm k will be found suboptimal with probability at least $1 - \mathcal{O}\left(\frac{NM}{T}\right)$ after $t_k = \mathcal{O}\left(\frac{\log(T)}{\Delta_k^2}\right)$ exploration pulls (for a single player) without being found optimal by any player before. Since the exploration phases are collision-less, the cost for pulling the sub-optimal arm k is $\mathcal{O}\left(\min\left\{\frac{M\log(T)}{\Delta_k}, \Delta_k T\right\}\right)$.

The same reasoning as in Appendix A.2.3 shows that the exploration regret due to non pulls of optimal arms is in $\mathcal{O}\left(\sum_{k>M}\min\left\{\frac{M\log(T)}{\mu_{(M)}-\mu_{(k)}}, \sqrt{T\log(T)}\right\}\right)$ conditioned to the success of the estimations.

As $N = \mathcal{O}(\log(T))$, all those arguments yield the bound on R^{explo}, with probability $1 - \mathcal{O}\Big(\frac{KM\log(T)}{T} + KM\exp(-\mu_{\min}T_c) + (K + \log(T))\frac{KM}{T}\Big)$. □

Theorem 2 can now be deduced from Lemmas 5, 6, 7 and Equation (1).

C Complementary material for Section 4

C.1 Algorithm description

DYN-MMAB is described in Algorithm 3 below. An arm is now said **active** for player j if it was neither detected as occupied nor optimal by this player yet. Using the notations of Algorithm 3, this means that this arm is not in `Preferences` ∪ `Occupied`.

C.1.1 Exploration rules

A player can take two types of decision during the exploration phase. First, an arm can be considered as occupied and is then added to `Occupied`. Or it can be considered as next optimal arm and is then added to `Preferences`.

Algorithm 3 DYN-MMAB algorithm

Input: T^j (horizon), $B^j(t) = 2\sqrt{\frac{3\ K\log(T^j)}{t}}$ (error bound)

1: Initialize $p \leftarrow 1$ and `Preferences`, `Occupied` as empty sets
2: Initialize $\mathbf{T}, \mathbf{T}^{\text{temp}}, \mathbf{S}, \mathbf{S}^{\text{temp}} \leftarrow \text{Zeros}(K)$ and $\mathbf{L}$ as a vector of K elements equal to ∞
3: **Exploration Phase:**
4: **while** not fixed yet **do**
5: Pull $k \sim \mathcal{U}([K])$; $T^{\text{temp}}[k] \leftarrow T^{\text{temp}}[k] + 1$ and $T[k] \leftarrow T[k] + 1$
6: $S^{\text{temp}}[k] \leftarrow S^{\text{temp}}[k] + r_k(t)$ and $S[k] \leftarrow S[k] + r_k(t)$
7: Update confidence int. $r_{\inf}[k] \leftarrow \left(\frac{S[k]}{T[k]} - B^j(t)\right)^+$ and $r_{\sup}[k] \leftarrow \min\left(\frac{S[k]}{T[k]} + B^j(t), 1\right)$
8: $L[k] \leftarrow \min\left(\frac{e\log(T^j)}{(r_{\inf}[k])^+},\ L[k]\right)$
9: **if** $k =$ `Preferences`$[p]$ and $r_k(t) > 0$ **then** `Fixed` $\leftarrow k$ **end if** *# no collision on the arm to exploit*
10: **if** `Preferences`$[p] \in$ `Occupied` **then** $p \leftarrow p + 1$ **end if** *# k is exploited by another player*
11: **if** $T^{\text{temp}}[k] \geq L[k]$ **then** *# end of sliding window*
12: **if** $S^{\text{temp}}[k] = 0$ **then** Add k to `Occupied` **end if** *# estimate that k is exploited*
13: Reset $S^{\text{temp}}[k], T^{\text{temp}}[k] \leftarrow 0$
14: **end if**
15: **if** for some active arm i and all other active arms l, $r_{\inf}[i] > r_{\sup}[l]$ **then**
16: Add i to `Preferences` *# i is better than any other active arm*
17: **end if**
18: **if** there is some l not in `Preferences`$[0 : p]$, such that $r_{\inf}[l] > r_{\sup}[k]$, where $k =$ `Preferences`$[p]$ **then** Add `Preferences`$[p]$ to `Occupied`
19: **end if** *# there is a "suboptimal" arm better than the optimal*
20: **end while**
21: **Exploitation Phase:** Pull `Fixed` until T^j

An arm k is added to `Occupied` (it may already be in `Preferences`) if only 0 rewards have been observed during a whole block of L_k^j pulls on arm k for player j. Such a block ends when at least L_k^j new observations have been gathered on arm k and a new block is then restarted. L_k^j is an estimation of the required number of successive 0 to observe before considering an arm as occupied with high probability. Its value at stage $t + \tau_j$, $L_k^j(t)$, is thus constantly updated using the current estimation of a lower bound for μ_k:

$$L_k^j(t+1) \leftarrow \min\left(\frac{e\log(T^j)}{\left(\hat{r}_k^j(t+1) - B^j(t+1)\right)^+},\ L_k^j(t)\right) \quad \text{and } L_k^j(0) = +\infty \tag{10}$$

where $\hat{r}_k^j(t)$ and $T_k^j(t)$ are respectively the empirical mean reward and number of pulls of player j on the arm k at stage $t + \tau_j$, $B^j(t) = 2\sqrt{\frac{3\ K\log(T^j)}{t}}$, $x^+ = \max(x, 0)$ and $\frac{e\log(T^j)}{0} = +\infty$. This rule is described at lines 11-14 in Algorithm 3.

An active arm k is added to `Preferences` if it is better, with high probability, than any other active arm, in term of confidence interval. This interval is only updated for the pulled arm at each stage. This rule is described at lines 15-17 in Algorithm 3.

Another rule needs to be added to handle the possible case of an arm in `Preferences` already exploited by another player: as soon as an arm k in `Preferences` becomes worse (in terms of confidence intervals) than an active arm or an arm with a higher index in `Preferences`, then k is added to `Occupied`. This rule is described at lines 18-19 in Algorithm 3.

Following these rules, as soon as there is an arm in `Preferences`, player j tries to occupy the p-th arm in `Preferences` (starting with $p = 1$), yet she still continues to explore. As soon as she encounters a positive reward, she occupies it and starts the exploitation phase. If she does not end occupying an optimal arm, this arm will end in `Occupied` at some point. The player then increments p and tries to occupy the next optimal arm. This point is described at lines 9-10 in Algorithm 3. Notice that `Preferences` can have more than p elements, but the player must not try to exploit the q-th element of `Preferences` with $q > p$ yet as it can lead the player in exploiting a sub-optimal arm.

C.2 Theoretical Analysis

This section is devoted to the proof of Theorem 3. It first proves several required lemmas.

To handle the fact that players can enter the game at anytime, we now introduce the quantity $\gamma^j(t)$, the expected multiplicative factor of the means defined by:

$$\gamma^j(t) = \frac{1}{t} \sum_{t'=\tau_j+1}^{t+\tau_j} \mathbb{E}\Big[(1 - \frac{1}{K})^{M_{t'}-1}\Big]$$

where M_t is the number of **active players** (i.e., sampling uniformly at random) at time t. A crucial point is that $\gamma^j(t) \geq \frac{1}{e}$, since $M \leq K$. As it is uniformly bounded away from 0, estimating $\gamma^j(t)\mu_k$ takes roughly the same time (up to a multiplicative constant) than estimating μ_k.

Lemma 8. *For any player j and arm k:*

1. *if k is still free at stage $t + \tau_j$, then $\mathbb{P}\Big[|\hat{r}_k^j(t) - \gamma^j(t)\mu_k| > 2\sqrt{\frac{3\ K\log(T^j)}{t}}\Big] \leq \frac{4}{T^j}$.*
2. *On the other hand, if k is occupied by some player $j' \neq j$ at stage $t^0+\tau_j$, then, conditionally to the correct estimation of all the arms until the considered stage, with probability $1 - \mathcal{O}\left(\frac{1}{T^j}\right)$:*
   - *either k is detected as "occupied" at a stage at most $t^0 + \tau_j + \mathcal{O}\left(\frac{K\log(T)}{\mu_k}\right)$ by player j*
   - *or, with $t_i = \mathcal{O}\left(\frac{K\log(T)}{\mu_i^2}\right)$, for some other arm i such that $\mu_i \leq \mu_k$ and still free at stage $t_i + \tau_j$, arm k is detected as sub-optimal and dominated by arm i for player j after a stage at most $t_i + \tau_j$*

Proof. 1. We first denote $Z_t = X_k(t+\tau_j)(1-\eta_k(t+\tau_j))\mathbb{1}_{\pi^j(t+\tau_j)=k}$ and $(\mathcal{F}_{t'})_{t'\in[t]}$ the filtration associated to the stages $\{1+\tau_j, \dots, t+\tau_j\}$. We define p_t such that $Z_t|\mathcal{F}_{t-1} \sim$ Bernoulli(p_t). Chernoff bound states that:

$$\mathbb{P}\Big[\sum_{t'=1}^{t}(Z_{t'} - \mathbb{E}[Z_{t'}]) \geq t\delta\Big] \leq \min_{\lambda>0} e^{-\lambda t\delta}\mathbb{E}\big[\prod_{t'=1}^{t} e^{\lambda(Z_{t'}-\mathbb{E}[Z_{t'}])}\big]$$

We can condition the expectation to $\mathcal{F}_{t-1}$. It then holds:

$$\mathbb{E}\Big[e^{\lambda(Z_t-\mathbb{E}[Z_t])}\,|\mathcal{F}_{t-1}\Big] = e^{-\lambda p_t}\left(1 + p_t(e^\lambda - 1)\right) \leq e^{-\lambda p_t}e^{p_t(e^\lambda-1)} \qquad \text{as } 1 + x \leq e^x$$

$$\leq e^{p_t(e^\lambda - 1 - \lambda)} \leq e^{\frac{e^\lambda - 1 - \lambda}{K}} \qquad \text{as } p_t \leq \frac{1}{K} \text{ and } e^\lambda - 1 - \lambda \geq 0$$

It can then be deduced by induction:

$$\mathbb{P}\Big[\sum_{t'=1}^{t}(Z_{t'} - \mathbb{E}[Z_{t'}]) \geq t\delta\Big] \leq \min_{\lambda>0} e^{-\lambda t\delta} e^{t\frac{e^\lambda - 1 - \lambda}{K}}. \qquad \text{For } \lambda = \log(1+K\delta):$$

$$\leq \exp\left(-\frac{t}{K}h(K\delta)\right) \qquad \text{with } h(u) = u\log(1+u) - u$$

Similarly, we show for the negative error: $\mathbb{P}\Big[\sum_{t'=1}^{t}(Z_{t'} - \mathbb{E}[Z_{t'}]) \leq -t\delta\Big] \leq \exp\big(-\frac{t}{K}h(-K\delta)\big)$.

Either $t \leq \frac{8}{3}K\log(T^j)$ and the desired inequality holds almost surely, or $K\delta < 1$ with $\delta = \sqrt{\frac{8\log(T^j)}{3tK}}$. As $h(x) \geq \frac{3x^2}{8}$ for $|x| < 1$, it then holds:

$$\mathbb{P}\Big[\Big|\sum_{t'=1}^{t}(Z_{t'} - \mathbb{E}[Z_{t'}])\Big| \geq t\delta\Big] \leq 2e^{-\frac{3t(K\delta)^2}{8K}} \qquad \text{and after multiplication with } \frac{K}{t}:$$

$$\mathbb{P}\left[\Big|\frac{K}{t}\sum_{t'=1+\tau_j}^{t+\tau_j} X_k(t')(1-\eta_k(t'))\mathbb{1}_{\pi^j(t')=k} - \gamma_j(t)\mu_k\Big| \geq \sqrt{\frac{8K\log(T^j)}{3t}}\right] \leq \frac{2}{T^j} \tag{11}$$

Second, Chernoff bound provides a confidence interval on the number of pulls on a single arm:

$$\mathbb{P}\left[\Big|T_k^j(t) - \frac{t}{K}\Big| \geq \sqrt{\frac{3t\log(T^j)}{K}}\right] \leq \frac{2}{T^j} \tag{12}$$

From Equation (12), it can be directly deduced that $\mathbb{P}\Big[|\frac{KT_k^j(t)}{t} - 1| \geq \sqrt{\frac{3K\log(T^j)}{t}}\Big] \leq \frac{2}{T^j}$. As $\hat{r}_k^j(t) \leq 1$:

$$\mathbb{P}\left[\Big|\frac{KT_k^j(t)}{t}\hat{r}_k^j(t) - \hat{r}_k^j(t)\Big| \geq \sqrt{\frac{3K\log(T^j)}{t}}\right] \leq \frac{2}{T^j} \tag{13}$$

As $\frac{K}{t}\sum_{t'=1+\tau_j}^{t+\tau_j} X_k(t')(1-\eta_k(t'))\mathbb{1}_{\pi^j(t')=k} = \frac{KT_k^j(t)}{t}\hat{r}_k^j(t)$, using the triangle inequality with Equations (11) and (13) finally yields $\mathbb{P}\Big[|\hat{r}_k^j(t) - \gamma^j(t)\mu_k| \geq 2\sqrt{\frac{3\ K\log(T^j)}{t}}\Big] \leq \frac{4}{T^j}$.

2. Since player j only updates the confidence bound of a single arm at each time step, the previous point gives that with probability $1 - \mathcal{O}\left(\frac{t}{T^j}\right)$, she correctly estimated all the free arms until stage $t + \tau_j$. The remaining of the proof is conditioned to this event until the last considered stage in the proof. We also assume that t^0 is the first stage where k is occupied for the proof. The general result claimed in Lemma 8 directly follows.

a) We first assume $t^0 \leq 12K\log(T^j)$. The empirical reward after $T_k^j(t) \geq T_k^j(t^0)$ pulls is $\hat{r}_k^j(t) = \frac{\hat{r}_k^j(t^0)T_k^j(t^0)}{T_k^j(t)}$, because all pulls after the stage $t^0 + \tau_j$ will return 0 rewards. However, using Equation (12), it appears that if $t^0 \leq 12K\log(T^j)$ then $T_k^j(t^0) \leq 18\log(T^j)$ with probability $1 - \mathcal{O}\left(\frac{1}{T^j}\right)$, so $\hat{r}_k^j(t) \leq \frac{18\log(T^j)}{T_k^j(t)}$.

Therefore with $t_i^j = \frac{72Ke\log(T^j)}{\mu_i^2}$, since $t_i \geq 12K\log(T^j)$, Chernoff bound guarantees that the following holds, with probability at least $1 - \frac{2}{T^j}$:

$$\frac{3t_i^j}{2K} \geq T_k^j(t_i) \geq \frac{t_i^j}{2K} = \frac{36e\log(T^j)}{\mu_i^2} \tag{14}$$

which gives $\hat{r}_k^j(t_i^j) \leq \frac{\mu_i}{2e}$. After stage $\tau_j + \frac{d'K\log(T^j)}{\mu_i^2}$, where d' is some universal constant, the error bounds of both arms are upper bounded by $\frac{\mu_i}{8e}$. The confidence intervals would then be disjoint for the arms k and i. So k will be detected as worse than i after a time at most $\mathcal{O}\left(\frac{K\log(T)}{\mu_i^2}\right)$ as $T^j \leq T$.

b) We now assume that $12K\log(T^j) \leq t^0 \leq \frac{12\lambda K\log(T^j)}{\mu_k^2}$ with $\lambda = 16e^2$. It still holds $\hat{r}_k^j(t) = \frac{\hat{r}_k^j(t^0)T_k^j(t^0)}{T_k^j(t)}$. Correct estimations of the free arms are assumed in this proof, so in particular:

$$\hat{r}_k^j(t) \leq \frac{(\mu_k + B^j(t^0))T_k^j(t^0)}{T_k^j(t)} \tag{15}$$

From Equation (14), it holds that $T_k^j(t^0) \leq \frac{3t^0}{2K}$ with probability $1 - \mathcal{O}\left(\frac{1}{T^j}\right)$ and thus $B^j(t^0) \leq 3\sqrt{\frac{2\log(T^j)}{T_k^j(t^0)}}$. Also, $T_k^j(t) \geq \frac{d\log(T^j)}{2\mu_i\mu_k}$ for $t = d\frac{K\log(T^j)}{\mu_i^2}$. Equation (15) then becomes:

$$\hat{r}_k^j(t) \leq \frac{\mu_k T_k^j(t^0)}{T_k^j(t)} + \frac{B^j(t^0)T_k^j(t^0)}{T_k^j(t)} \leq \frac{36\lambda}{d}\mu_i + \frac{3\sqrt{2T_k^j(t^0)\log(T^j)}}{T_k^j(t)} \leq (\frac{36\lambda}{d} + \frac{36\sqrt{\lambda}}{d})\mu_i$$

Thus, for a well chosen d, the empirical reward verifies $\hat{r}_k^j(t) \leq \frac{\mu_i}{2e}$. We then conclude as for the first case that the arm k would be detected as worse than the free arm i after a time $\mathcal{O}\left(\frac{K\log(T)}{\mu_i^2}\right)$.

c) The last case corresponds to $t^0 > \frac{12\lambda K\log(T^j)}{\mu_k^2}$. It then holds $B^j(t^0) \leq \frac{\mu_k}{\sqrt{\lambda}} = \frac{\mu_k}{4e}$.

By definition, $L_k^j \leq \frac{e\log(T^j)}{\hat{r}_k^j - B^j(t)}$. Conditionally to the correct estimation of the free arms, it holds that $r_k^j - 2B^j(t) \leq \hat{r}_k^j - B^j(t) \leq r_k^j$. So with the choice of L_k^j described by Equation (10), as long as k is free:

$$\begin{aligned} \frac{e\log(T^j)}{r_k^j} \leq L_k^j \quad &\leq \frac{e\log(T^j)}{r_k^j - 2B^j(t)} \\ &\leq \frac{e^2\log(T^j)}{\mu_k - 2eB^j(t)} \end{aligned} \tag{16}$$

As $B^j(t^0) \leq \frac{\mu_k}{4e}$, it holds that $L_k^j(t^0) \leq \frac{2e^2\log(T^j)}{\mu_k}$. Since L_k^j is non-increasing by definition, this actually always holds for any t larger than t^0.

From that point, Equation (14) gives that with probability $1 - \mathcal{O}\left(\frac{1}{T^j}\right)$, the arm k will be pulled at least $2L_k^j$ times between stage $t^0 + 1$ and $t^0 + \max(4KL_k^j, 24K\log(T^j))$. Thus, a whole block of L_k pulls receiving only 0 rewards on k happens before stage $t^0 + \max(4KL_k^j, 24K\log(T^j))$.

The arm k is then detected as occupied after a time $\mathcal{O}\left(\frac{K\log(T^j)}{\mu_k}\right)$ from t^0, leading to the result. □

The second part of Lemma 8 states that an occupied arm will either be quickly detected as occupied (first case), or its average reward will significantly drop, so that it will be detected as sub-optimal after a certain number of pulls (second case). Thanks to the third rule presented in Appendix C.1.1, in the second case, even if the arm was previously detected as optimal, it will be removed from the set `Preferences`. Lemma 9 states that there will be, with high probability, no false detection of occupied arms.

Lemma 9. *No free arm k at stage $t + \tau_j$ was falsely detected as occupied in all the previous time steps by player j with probability $1 - \mathcal{O}\left(\frac{t}{T^j}\right)$.*

Proof. As shown above, with probability $1 - \mathcal{O}\left(\frac{t}{T^j}\right)$, player j correctly estimated the average rewards of all the free arms until stage $t + \tau_j$. The remaining of the proof is conditioned to that event. Equation (16) then gives that

$$L_k^j(t') \geq \frac{e \log(T^j)}{\mu_k} \qquad \text{at each stage } t' \leq t. \tag{17}$$

As in Appendix A.2.1, the probability of detecting L successive 0 rewards on a free arm k verifies:

$$\begin{aligned} \mathbb{P}\left[\forall s \in [L], r_k(s) = 0 \mid k \text{ free}\right] &\leq \left(\mathbb{P}\left[r_k(s) = 0\right]\right)^L \\ &\leq \left(1 - \frac{\mu_k}{e}\right)^L \leq \exp\left(-\frac{L\mu_k}{e}\right) \end{aligned} \tag{18}$$

The last line of Equation (18) along with Equation (17) provides that with probability $1-\mathcal{O}\left(\frac{1}{T^j}\right)$, at least one positive reward will be observed on arm k in L_k successive pulls of a single block. The union bound over all blocks yields the result. □

Finally, Lemma 10 yields that, after some time, any player starts exploiting an arm and all the better arms are already occupied by other players.

Lemma 10. *We denote $\bar{\Delta}_{(k)} = \min_{i=1,\dots,k}(\mu_{(i)} - \mu_{(i+1)})$. With probability $1 - \mathcal{O}\left(\frac{\bar{t}_M}{T^j}\right)$, it holds that for a single player j, there exists k_j such that after a stage at most $\bar{t}_{k_j} + \tau_j$, she is exploiting the k_j-th best arm and all the better arms are also exploited by other players, where $\bar{t}_{k_j} = \mathcal{O}\left(\frac{K\log(T)}{\bar{\Delta}^2_{(k_j)}} + k_j \frac{K\log(T)}{\mu_{(k_j)}}\right)$.*

Proof. Let us consider the player j. Player j correctly estimates all the arms until stage $\bar{t}_M + \tau_j$, with probability $1 - \mathcal{O}\left(\frac{\bar{t}_M}{T^j}\right)$. The remaining of the proof is conditioned to that event. We define $\bar{t}_i = \frac{cK\log(T^j)}{\bar{\Delta}^2_{(i)}} + i\frac{cK\log(T^j)}{\mu_{(i)}}$ for some universal constant c and k_j (random variable) as:

$$k_j = \min\{i \in [K] \mid i\text{-th best arm not exploited by another player at stage } \bar{t}_i + \tau_j\} \tag{19}$$

k_j^* (k^j-th best arm) is the best arm not exploited by another player (than player j) after the stage $\bar{t}_{k_j} + \tau_j$. The considered set is not empty as $M \leq K$.

Lemma 9 gives that with probability $1 - \mathcal{O}\left(\frac{\bar{t}_M}{T^j}\right)$, k_j^* is not falsely detected as occupied until stage $\bar{t}_M$. All arms below k_j^* will be detected as worse than k_j^* after a time $\frac{dK\log(T^j)}{\bar{\Delta}^2_{(k_j)}}$ for some universal constant d, as we assume correct estimations of the arms.

By definition of k_j, any arm i^* better than k_j^* is already occupied at stage $\bar{t}_i+\tau_j$. Lemma 8, gives that with probability $1-\mathcal{O}\left(\frac{1}{T^j}\right)$, either i^* is detected as occupied after stage $\bar{t}_i+\tau_j+\frac{d'K\log(T^j)}{\mu_{(i)}}$ or dominated by k_j^* after stage $\frac{d_2K\log(T^j)}{\bar{\Delta}^2_{(k_j)}}+\tau_j$ for some universal constants d' and d_2.

Thus the player detects the arm k_j^* as optimal and starts trying to occupy k_j^* at a stage at most $\tilde{t}=\max\left(\bar{t}_{k_j-1}+\frac{d'K\log(T^j)}{\mu_{(k_j)}},\max(d,d_2)\frac{K\log(T^j)}{\bar{\Delta}^2_{(k_j)}}\right)+\tau_j$ with $\bar{t}_0=1$.

Using similar arguments as for Lemma 9, player j will observe a positive reward on k_j^* with probability $1-\mathcal{O}\left(\frac{1}{T^j}\right)$ after a stage at most $\tilde{t}+\frac{d_2'K\log(T^j)}{\mu_{(k_j)}}$, if k_j is still free at this stage. With the choice $c=\max(d,d_2,d'+d_2')$, this stage is smaller than $\bar{t}_{k_j}$ and k_j^* is then still free. Thus, player j will start exploiting k_j^* after stage at most $\bar{t}_{k_j}$. □

C.2.1 Regret in Dynamic Setting

We now prove Theorem 3.

Proof. Lemma 10 states that a player only needs an exploration time bounded as $\mathcal{O}\left(\frac{K\log(T)}{\bar{\Delta}^2_{(k)}}+k\frac{K\log(T)}{\mu_{(k)}}\right)$ before starting exploiting, with high probability. Furthermore, the better arms are already exploited when she does so. Thus, the exploited arms are the top-M arms. Moreover, the regret is upper bounded by twice the sum of exploration times (and the low probability events of wrong estimations), as a collision between players can only happen with at most one of these players not in exploration phase.

The regret incurred by low probability events mentioned in Lemma 10 is also dominated by this term when summing over all players and using the same argument of collisions. □

D On the inefficiency of UCB

A linear regret for the UCB algorithm in the No Sensing model has been conjectured recently [Besson and Kaufmann, 2018]. This algorithm seems to have good results in practice, although rare runs with linear regret appear. This is due to the fact that with probability $p>0$ at some point t, both independent from T, some players might have the same number of pulls and the same observed average rewards for each arm. In that case, the players would pull the exact same arms and thus collide until they reach a tie breaking point where they could choose different arms thanks to a random tie breaking rule. However, it was observed that such tie breaking points would not appear in the experiments, explaining the linear regret for some runs. Here we claim that such tie breaking points might never happen in theory for the UCB algorithm, if we add the constraint that the numbers of positive rewards observed for the arms are all different at some stage. This event remains probable even with this additional constraint.

Theorem 4. *For $s,s'\in\mathbb{N}$ with $s\neq s'$, we have $\forall n\geq 2, t,t'\in\mathbb{N}$:*

$$\frac{s}{t}+\sqrt{\frac{2\log(n)}{t}}\neq\frac{s'}{t'}+\sqrt{\frac{2\log(n)}{t'}}$$

Proof. First, if $t=t'$, these two quantities are obviously different as $s\neq s'$.

We now assume $\frac{s}{t}+\sqrt{\frac{2\log(n)}{t}}=\frac{s'}{t'}+\sqrt{\frac{2\log(n)}{t'}}$ with $t\neq t'$.

This means $\sqrt{\frac{2\log(n)}{t}} - \sqrt{\frac{2\log(n)}{t'}} \in \mathbb{Q}$, i.e., for some $p \in \mathbb{Q}$, it holds that $\log(n)(t + t' - 2\sqrt{tt'}) = 2p$.

It then holds

$$\log(n)\sqrt{tt'} = \log(n)\frac{t+t'}{2} - p$$

$$tt'\log^2(n) = \log^2(n)(\frac{t+t'}{2})^2 - p(t+t')\log(n) + p^2$$

$$\log^2(n)(\frac{t-t'}{2})^2 - p(t+t')\log(n) + p^2 = 0$$

Since $(\frac{t-t'}{2})^2 \neq 0$ and all the coefficients are in $\mathbb{Q}$ here, this would mean that $\log(n)$ is an algebraic number. However, Lindemann–Weierstrass theorem implies that $\log(n)$ is transcendental for any integer $n \geq 2$. We thus have a contradiction. □

The proof is only theoretical as computer are not precise enough to distinguish rationals from irrationals. The advanced arguments are not applicable in practice. Still, this seems to confirm the conjecture proposed by Besson and Kaufmann [2018]: there is not any tie breaking point, or at least for a very long period of time.